Chapter 3

Radioactivity

3.1 INTRODUCTION

In the periodic table there are nuclei, which are called stable, i.e. the properties of these nuclei do not change with the passage of time, unless the nuclei are otherwise disturbed. Yet there is another class of nuclei, whose number is much more, which are unstable. They disintegrate spontaneously by emitting either electromagnetic radiations or some particles.

The phenomenon of spontaneous decay of a nucleus accompanied by the emission of alpha-particles, beta-particles or gamma-rays is known as radioactivity.

It could be either natural or artificial. Radioactivity was first discovered by Henry Becquerel in 1896 accidentally in the same way as X-rays were discovered by Röentgen in 1895. French scientist Henry Becquerel in 1896 accidentally discovered the phenomenon of radioactivity, when he took a photographic plate wrapped in black paper and placed a uranium salt on it and exposed it to sunlight. On developing the photographic plate he found it to be foggy or darkened indicating that the uranium salt has emitted radiations in the presence of sunlight, which could penetrate into the black paper and affect the photographic plate. In those days it was thought that this phenomenon was possible only in sunlight. Once again Becquerel got ready to repeat this experiment, but the day was cloudy. So he put the whole set-up into his table drawer. After two days, he developed these plates without exposing to sun. To his surprise, the photographic plates were still foggy. It became clear to him that the uranium salt always gives invisible radiations irrespective of illumination. He was able to show that these invisible penetrating radiations are able to ionize gases and some part of the radiations consisted of fast moving charged particles. This was the beginning of the era of radioactivity.

Soon afterwards, Pierre Curie and Marie Curie working in the Becquerel laboratory, found two elements other than Uranium which were also radioactive. They named one as Polonium and the other as Radium.

The experiments of Rutherford and his co-workers found out three components in these penetrating radiations. They called these components as *a*-, *b*- and *g*-particles. Later on *a*-particles were identified as ^{4}He nuclei, *b*-particles as electrons and *g*-particles as high-energy photons. Later studies discovered three modes of *b*-decay. They are electron emission (b^{-}), positron emission (b^{+}) and electron capture. Similarly, *g*-decay also has three different decay modes. They are *g*-emission, internal conversion and internal pair conversion. The various decay modes of a radioactive nucleus are summarized in Figure 3.1. These modes are given below:

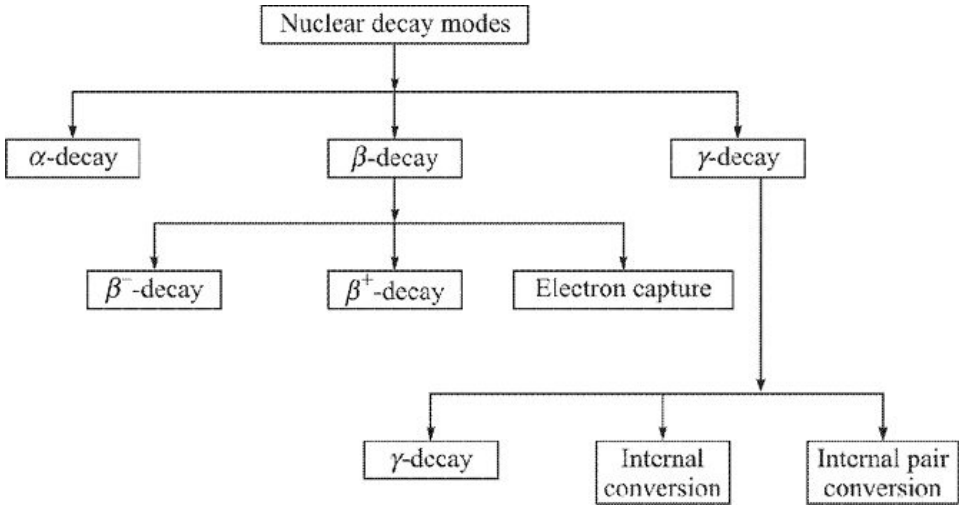


Figure 3.1 Various decay modes of radioactive nuclei.

Alpha Decay

In this process, parent nucleus disintegrates to give a daughter nucleus and helium nucleus or an *a*-particle. Mass number of the daughter nucleus decreases by four units and atomic number decreases by two units. A typical example of this decay mode is

$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

Beta Decay

In this process, parent nucleus disintegrates to give a daughter nucleus and a *b*-particle. As shown in Figure 3.1, *b*-decay is classified into three categories.

(i) **Electron emission or b^--decay:** Here, parent nucleus disintegrates to give a daughter nucleus, a b^--particle (or an electron) and a new particle called *antineutrino* $(\bar{n})$. Mass number of the daughter nucleus remains the same and atomic number increases by one unit. One such example is

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + \beta^- + \bar{\nu}$$

(ii) **Positron emission or b^+-decay:** In this process, parent nucleus disintegrates to give a daughter nucleus, a b^+-particle (or a positron) and a new particle called *neutrino* (*n*). Mass number of the daughter nucleus remains the same and atomic number decreases by one unit. One example is

$$^{22}_{11}\mathrm{Na} \rightarrow {}^{22}_{10}\mathrm{Ne} + \beta^+ + \nu$$

(iii) **Electron capture:** Here the parent nucleus captures one of the orbital electrons with the emission of a neutrino (*n*). Mass number of the daughter nucleus remains the same and atomic number decreases by one unit. For example,

$$^{54}_{25}\mathrm{Mn} + \beta^- \rightarrow {}^{54}_{24}\mathrm{Cr} + \nu$$

Gamma Decay

If the excitation energy available with a nucleus is not sufficient for particle emission, it loses its energy by following three processes, as shown in Figure 3.1.

(i) **Gamma decay:** Alpha and beta decays of a radioactive nucleus usually leave the daughter nucleus in an excited state. If the excitation energy available with the daughter nucleus is not sufficient for further particle emission, it loses its energy by emitting electromagnetic radiations, also known as *g*-rays. Mass and charge of the daughter nucleus remains the same as before the emission of *g*-rays. One such example is

$$^{137}_{56}\mathrm{Ba}^* \rightarrow {}^{137}_{56}\mathrm{Ba} + \gamma$$

The star (*) on ^{137}Ba indicates that it is in excited state.

(ii) **Internal conversion:** In internal conversion process, an excited nucleus instead of emitting *g*-rays directly transfers its excitation energy to one of the orbital electrons and the orbital electron is ejected as conversion electron. Mass and charge of the daughter nucleus remains the same as before the emission of the orbital electron. This is so because electron is ejected from the electronic orbits.

(iii) **Internal pair conversion:** If the excitation energy of the nucleus > 1.022 MeV, the nucleus can de-excite by emitting directly an electron and positron pair in its own Coulomb field. This process is known as *internal pair conversion*.

More details of these processes will be given later on.

3.2 LAWS OF DISINTEGRATION

The laws of radioactive disintegration are:

1. There is an equal probability for all nuclei of a radioactive element to decay.
2. The rate of spontaneous disintegration of a radioactive element is proportional to the number of nuclei present at that time.

Mathematically, it can be written as

$$\frac{dN}{dt} \propto N \tag{3.1}$$

where N is the number of atoms present at time t. Removing proportionality sign, we get

$$\frac{dN}{dt} = -\lambda N \tag{3.2}$$

where l is a constant of proportionality and is known as decay constant of the element. Negative sign indicates that as t increases N decreases. Rewriting Eq. (3.2) as

$$\frac{dN}{N} = -\lambda dt \tag{3.3}$$

Integrating both sides, we have

$$\int \frac{dN}{N} = -\lambda \int dt$$

$$\ln(N) = -lt + C \qquad (3.4)$$

where C is constant of integration and is evaluated by the fact that at $t = 0$, number of atoms of the radioactive element is N_0. Using this condition, we get

$$C = \ln(N_0) \qquad (3.5)$$

Substituting this value of C in Eq. (3.4), we get

$$\ln(N) = -\lambda t + \ln(N_0)$$

or

$$\ln \frac{N}{N_0} = -\lambda t$$

Thus,

$$\boxed{N = N_0 e^{-\lambda t}} \qquad (3.6)$$

The exponential nature of this equation shows that it takes infinite time for whole of the radioactive material to disintegrate. N vs. t has been plotted for ^{24}Na($t_{1/2}$ = 15 h) in Figure 3.2. A plot of log(N) vs. t would be a straight line.

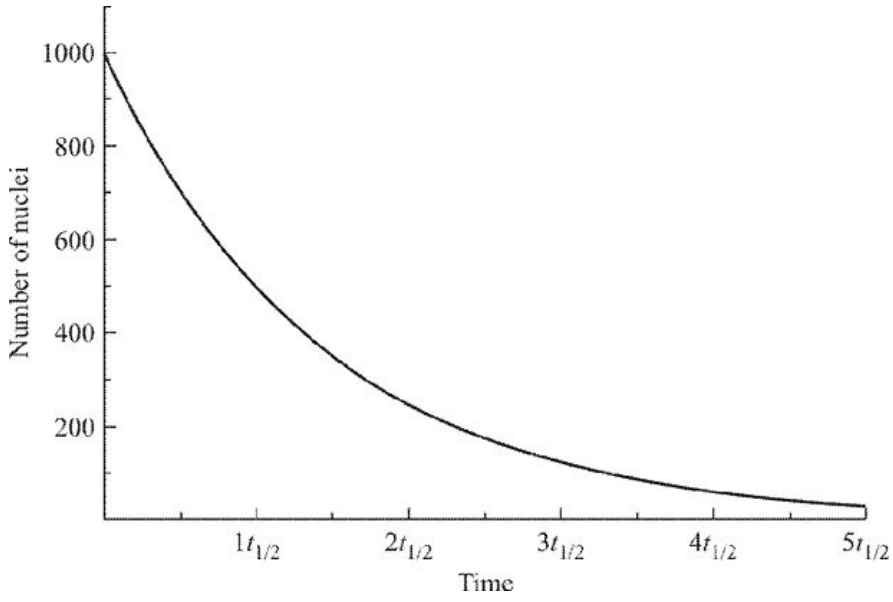


Figure 3.2 Exponential decay curve for ^{24}Na.

3.2.1 Activity and its Units

Activity

The activity (A) of a radioactive nuclide is defined as the number of disintegration per unit time.

$$A = -\frac{dN}{dt} = \lambda N$$

Substituting for N from Eq. (3.6), we get

$$A = lN_0e^{-l^t}$$

Substituting $A_0 = lN_0$, the exponential relation for activity is

$$\boxed{A = A_0e^{-\lambda t}} \qquad (3.7)$$

In this relation A_0 is the activity at $t = 0$. The exponential factor shows that activity is decreasing with time in the same fashion as N.

Unit

- **The SI unit of activity is becquerel (Bq):** 1 becquerel = 1 Bq = 1 disintegration/s (dps or disintegrations per second).

In day-to-day life, we encounter activities of the order of kilobecquerel (kBq), megabecquerel (MBq) or gigabecquerel (GBq).

$$1 \text{ kBq} = 10^3 \text{ Bq}$$

$$1 \text{ MBq} = 10^6 \text{ Bq}$$

$$1 \text{ GBq} = 10^9 \text{ Bq}$$

- **Traditional unit of activity is curie (Ci) and rutherford (R):**

$$1 \text{ curie (Ci)} = 3.7 \square 10^{10} \text{ disintegration/s}$$

$$= 3.7 \square 10^{10} \text{ Bq}$$

$$= 37 \text{ GBq}$$

$$1 \text{ rutherford} = 10^6 \text{ disintegration/s}$$

$$= 1 \text{ MBq}$$

Curie is a big unit, normal activities of radioactive sources are defined in terms of millicurie (1 mCi = 10^{-3} Ci) and microcurie (1 □Ci = 10^{-6} Ci).

3.2.2 Half-life

Rutherford in 1904 introduced the term *half-life* ($t_{1/2}$) for a radioactive material. It is the characteristic property of a radioactive nuclide.

Half-life is the time required for number of radioactive nuclides to reduce to half by disintegration.

If N_0 is the initial number of radioactive nuclei present, then in one half-life this number will reduce to $N_0/2$. Therefore, after 1 half-life

$$N = \frac{N_0}{2}$$

Substituting N in Eq. (3.6), we get

$$\frac{N_0}{2} = N_0 e^{-\lambda t_{1/2}}$$

or

$$\frac{1}{2} = e^{-\lambda t_{1/2}}$$

or

$$e^{\lambda t_{1/2}} = 2$$

Taking natural log on both sides

$$\lambda t_{1/2} = \ln 2$$

$$\boxed{t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}} \quad (3.$$

Therefore, half-life of a radioactive material depends upon the disintegration constant *l*. The larger the value of disintegration constant, smaller is the half-life, i.e. nuclei of radioactive material will decay faster and vice versa. Half-life of a radioactive material is its intrinsic property and it cannot be altered by any physical or chemical means.

After each half-life, the number of radioactive atoms reduces to half. For example, if we take on 1st Jan. 2006, 10,000 atoms of a radioactive material whose half-life is 1 year, then on 1st Jan. 2007, the number of radioactive atoms left becomes 5000. On 1st Jan. 2008, the number will be 2500 and on 1st Jan. 2009, it will be 1250 and so on. It is clear that after infinite half-lives, the number of radioactive nuclei will reduce to zero.

3.2.3 Average (Mean) Life

As we have seen (Eq. 3.6) that infinite time is required for the complete disappearance of the radioactive material, one does not know which atom of the material is going to decay at a given instant of time, i.e. individual atoms have lifetime from zero to infinity. Because of this Soddy in 1904 introduced the concept of average or mean life (t).

The average or mean life is defined as the average life expectancy of the radioactive species.

Therefore, the average life is calculated by summing the lives of all the nuclei and dividing by the total number of nuclei. Suppose dN_1 nuclei decay in time t_1, dN_2 nuclei decay in time t_2, dN_3 nuclei decay in time t_3 and so on, then the average life or mean life will be

$$\tau = \frac{t_1 dN_1 + t_2 dN_2 + t_3 dN_3 + \ldots}{dN_1 + dN_2 + dN_3 + \ldots}$$

This equation can be written in integral form as

$$\tau = \frac{\int_0^{N_0} t dN}{\int_0^{N_0} dN}$$

Since

$$\int_0^{N_0} dN = N_0$$

Therefore,

$$\tau = \frac{1}{N_0} \int_0^{N_0} t dN$$

where N_0 is the total number of radioactive atoms present initially.

Differentiating Eq. (3.6) with respect to t, we have

$$dN = -lN_0 e^{-l^t} dt$$

Now for $t = \square$, $N = 0$ and for $t = 0$, $N = N_0$

Therefore,

$$\tau = -\frac{1}{N_0}\int_{\infty}^{0} \lambda t N_0 e^{-\lambda t} dt$$

$$\tau = \lambda \int_{0}^{\infty} t e^{-\lambda t} dt \tag{3.9}$$

The integral in Eq. (3.9) can be evaluated by parts

$$\int_{0}^{\infty} t e^{-\lambda t} dt = \left[t \frac{e^{-\lambda t}}{-\lambda} \right]_{0}^{\infty} - \int_{0}^{\infty} 1 \frac{e^{-\lambda t}}{-\lambda} dt$$

$$= (0 - 0) - \left[\frac{1}{\lambda} \times \frac{e^{-\lambda t}}{-\lambda} \right]_{0}^{\infty}$$

$$= -\frac{1}{\lambda^2}(0 - 1) = \frac{1}{\lambda^2}$$

Therefore, the value of average life from Eq. (3.9) is

$$\boxed{\tau = \frac{1}{\lambda} = \frac{t_{1/2}}{0.693}} \tag{3.10}$$

where we have used the value of l from Eq. (3.8).

3.3 RADIOACTIVE SERIES

There are about 65 naturally occurring radioisotopes. Out of these about 42 radioactive nuclides have Z □ 82. These radioactive nuclides decay by *a*- or *b*-emission, changing Z and/or A until a stable nucleus is reached. *a*-decay changes A by 4 amu while *b*-decay does not change A, therefore, we have four independent decay chains with mass numbers $4n$, $4n + 1$, $4n + 2$ and $4n + 3$, where n is an integer. Any other change of mass is repetition of these four series, i.e. $4n + 4$ will be a member of $4n$ series, and so on. This decay process tends to terminate to the stable member of the series. These four series are listed in Table 3.1.

TABLE 3.1 Radioactive series

Series	Type	Parent nucleus	Final nucleus	Half-life of parent nucleus
Thorium	$4n$	^{232}Th	^{208}Pb	1.41 □ 10^{10}
Neptunium	$4n + 1$	^{237}Np	^{209}Bi	2.14 □ 10^{6}
Uranium	$4n + 2$	^{238}U	^{206}Pb	4.47 □ 10^{9}

Actinium	$4n + 3$	^{235}U	^{207}Pb	7.04×10^8

Notice that the neptunium series is not observed in nature as the longest-lived member of the neptunium series has far too short a lifetime (2.14×10^6 years) *to have survived since the formation of the earth* (~4.6×10^9 years). *Further, unlike other series, the parent nucleus of actinium series is* ^{235}U *and not actinium. This is due to the fact that actinium was first discovered and it was thought that actinium is the parent nucleus of this series. So, this series was called actinium series. But later on when* ^{235}U *was discovered, it was found that not actinium but long- lived* ^{235}U (7.04×10^8 years) *is the parent nucleus of this series. However, the name actinium series was not changed and still continuing.*

The four series corresponding to $4n$, $4n + 1$, $4n + 2$ and $4n + 3$ are displayed in Figures 3.3 to 3.6 respectively.

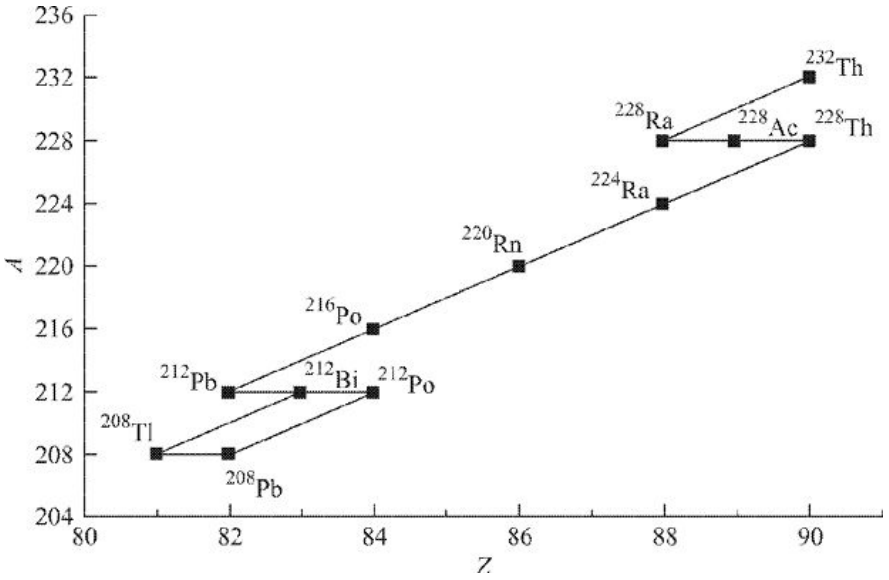


Figure 3.3 $A = 4n$, Thorium series. End product is ^{208}Pb.

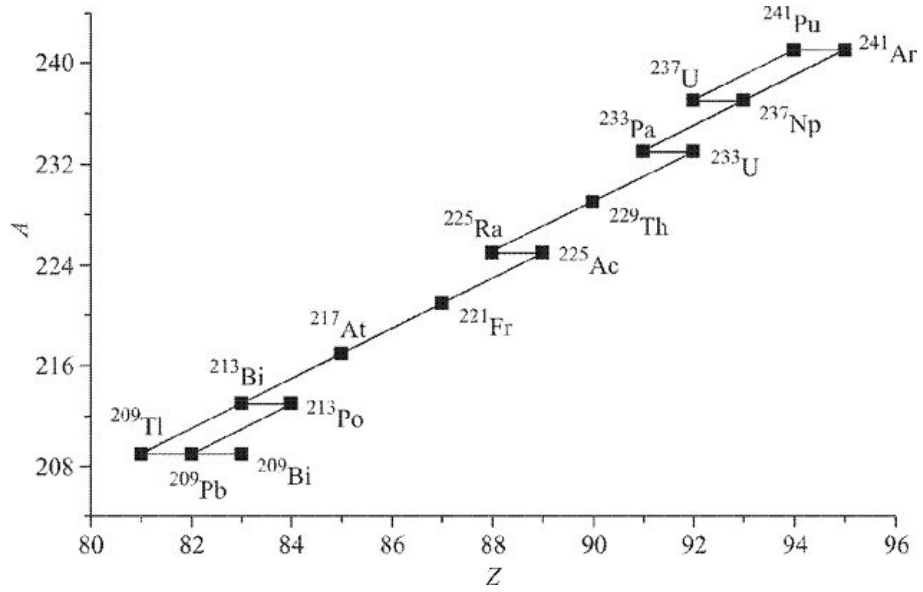


Figure 3.4 $A = 4n + 1$, Neptunium series. End product is ^{209}Bi.

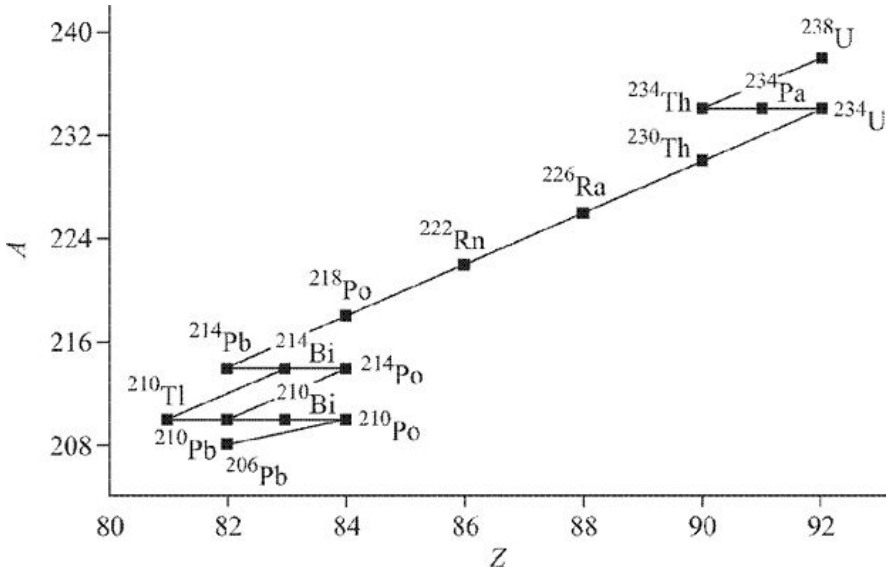


Figure 3.5 $A = 4n + 2$, Uranium series. End product is ^{206}Pb.

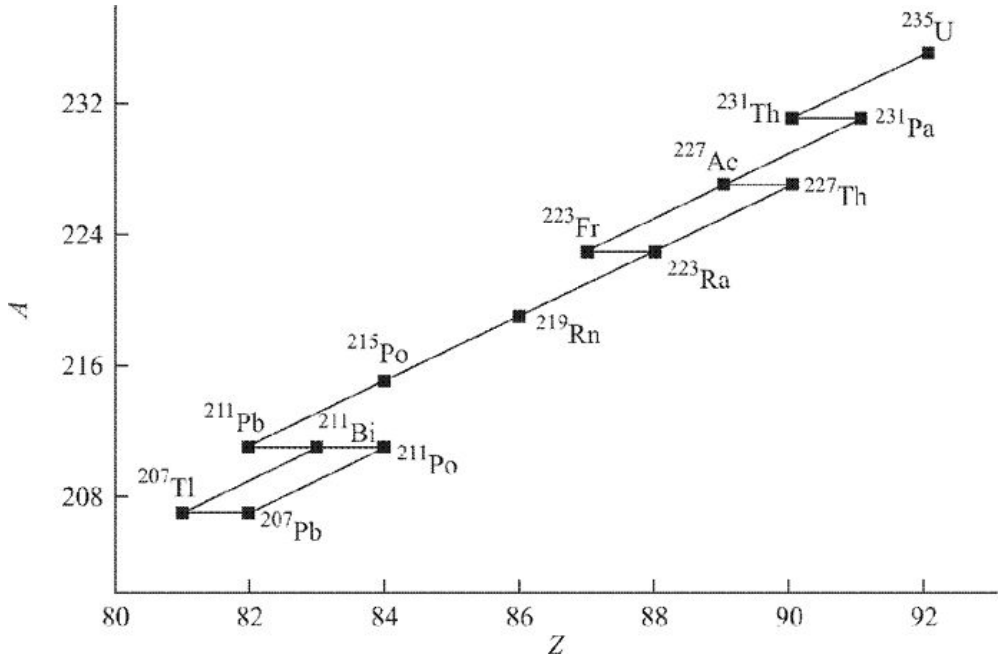


Figure 3.6 $A = 4n + 3$, Actinium series. End product is ^{207}Pb.

3.4 LAW OF SUCCESSIVE DISINTEGRATION

In all the examples considered so far, radioactive nucleus also known as parent nucleus decays to a stable nucleus, called daughter nucleus. There can be other cases too.

A parent nucleus decays into a daughter nucleus, which is also radioactive. The daughter nucleus further decays into another nucleus known as granddaughter. This series can continue till a stable nucleus is obtained. This whole process is known as *successive disintegrations* or a *radioactive series*.

Some typical examples are

$$^{105}_{44}\text{Ru} \xrightarrow{\beta^-} {}^{105}_{45}\text{Rh} \xrightarrow{\beta^-} {}^{105}_{46}\text{Pd (stable)}$$

$$^{140}_{54}\text{Xe} \xrightarrow{\beta^-} {}^{140}_{55}\text{Cs} \xrightarrow{\beta^-} {}^{140}_{56}\text{Ba} \xrightarrow{\beta^-} {}^{140}_{57}\text{La} \xrightarrow{\beta^-} {}^{140}_{58}\text{Ce (stable)}$$

An Interesting Question Arises

If we start with a given number of atoms of the parent radioactive nucleus, what is the number of nuclei in each of daughter and granddaughter at any given time?

At any time t, let N_1 be the number of atoms of the parent nucleus, which decays with decay constant l_1 into its daughter nucleus. Let N_2 be the

number of nuclei of the daughter element, which in turn decays with a decay constant l_2 into a stable element which has N_3 stable atoms. Also assume that at time $t = 0$, $N_1 = N_1{}^0$, $N_2 = N_2{}^0$ and $N_3 = N_3{}^0$.

Now, the rate of decay of the parent nucleus is

$$\frac{dN_1}{dt} = -\lambda_1 N_1 \tag{3.11}$$

The number of atoms of the daughter element increases at a rate $l_1 N_1$ and at the same time this number decreases at a rate $l_2 N_2$ per second. Hence, the rate of decay of the daughter element is

$$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2 \tag{3.12}$$

Finally, the growth of granddaughter is

$$\frac{dN_3}{dt} = \lambda_2 N_2 \tag{3.13}$$

Integrating Eq. (3.11) and imposing condition that $N_1 = N_1{}^0$ at $t = 0$, gives

$$\boxed{N_1 = N_1^0 e^{-\lambda_1 t}} \tag{3.14}$$

Substituting Eq. (3.14) in Eq. (3.12), we get

$$\frac{dN_2}{dt} = \lambda_1 N_1^0 e^{-\lambda_1 t} - \lambda_2 N_2$$

$$\frac{dN_2}{dt} + \lambda_2 N_2 = \lambda_1 N_1^0 e^{-\lambda_1 t} \tag{3.15}$$

Multiplying Eq. (3.15) by $e^{\lambda_2 t}$, we have

$$e^{\lambda_2 t}\frac{dN_2}{dt} + \lambda_2 N_2 e^{\lambda_2 t} = \lambda_1 N_1^0 e^{-\lambda_1 t} e^{\lambda_2 t}$$

or

$$\frac{d}{dt}[N_2 e^{\lambda_2 t}] = \lambda_1 N_1^0 e^{(\lambda_2 - \lambda_1)t} \tag{3.16}$$

Integrating Eq. (3.16), we get

$$N_2 e^{\lambda_2 t} = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 e^{(\lambda_2 - \lambda_1)t} + C \tag{3.17}$$

where C is a constant of integration. At $t = 0$, $N_2 = N_2^0$

$$C = -\frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 + N_2^0$$

Substituting C in Eq. (3.17), we get

$$N_2 e^{\lambda_2 t} = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 e^{(\lambda_2 - \lambda_1)t} - \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 + N_2^0$$

Dividing by $e^{\lambda_2 t}$, we get

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 e^{-\lambda_1 t} - \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 e^{-\lambda_2 t} + N_2^0 e^{-\lambda_2 t}$$

$$\boxed{N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 (e^{-\lambda_1 t} - e^{-\lambda_2 t}) + N_2^0 e^{-\lambda_2 t}} \tag{3.18}$$

Substituting for N_2 in Eq. (3.13) and integrating with respect to t, we get

$$N_3 = \lambda_2 \int \left\{ \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 [e^{-\lambda_1 t} - e^{-\lambda_2 t}] + N_2^0 e^{-\lambda_2 t} \right\} dt$$

$$N_3 = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} N_1^0 \left[\frac{e^{-\lambda_1 t}}{-\lambda_1} - \frac{e^{-\lambda_2 t}}{-\lambda_2} \right] + \lambda_2 N_2^0 \frac{e^{-\lambda_2 t}}{-\lambda_2} + D$$

where D is a constant of integration. Simplifying, we get

$$N_3 = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} N_1^0 \left[\frac{e^{-\lambda_2 t}}{\lambda_2} - \frac{e^{-\lambda_1 t}}{\lambda_1} \right] - N_2^0 e^{-\lambda_2 t} + D \tag{3.19}$$

At $t = 0$ $N_3 = N_3^0$, we get

$$D = N_1^0 + N_2^0 + N_3^0$$

Substituting D in Eq. (3.19), we get

$$N_3 = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} N_1^0 \left[\frac{e^{-\lambda_2 t}}{\lambda_2} - \frac{e^{-\lambda_1 t}}{\lambda_1} \right] - N_2^0 e^{-\lambda_2 t} + N_1^0 + N_2^0 + N_3^0$$

$$\boxed{N_3 = N_3^0 + N_2^0 (1 - e^{-\lambda_2 t}) + N_1^0 \left(1 + \frac{\lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} - \frac{\lambda_2}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} \right)} \tag{3.20}$$

Number of nuclei of parent N_1, daughter N_2 and granddaughter N_3 at any given instant of time t can be calculated using Eqs. (3.14), (3.18) and (3.20) respectively.

Special Case

When only parent nuclei are present at $t = 0$, i.e. $N_1 = N_1{}^0$ and
Substituting these values, Eqs. (3.14), (3.18) and (3.20) for $t > 0$ become

$$N_1 = N_1^0 e^{-\lambda_1 t} \tag{3.21}$$

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 (e^{-\lambda_1 t} - e^{-\lambda_2 t}) \tag{3.22}$$

and

$$N_3 = N_1^0 \left(1 + \frac{\lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} - \frac{\lambda_2}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} \right) \tag{3.23}$$

With the help of Eqs. (3.21), (3.22) and (3.23), we can compute the number of nuclei of parent, daughter and granddaughter (assuming granddaughter to be stable) at any instant of time. In Figure 3.7, we have plotted the number of nuclei of parent, daughter and granddaughter versus t assuming initially at $t = 0$, only parent nuclei exist.

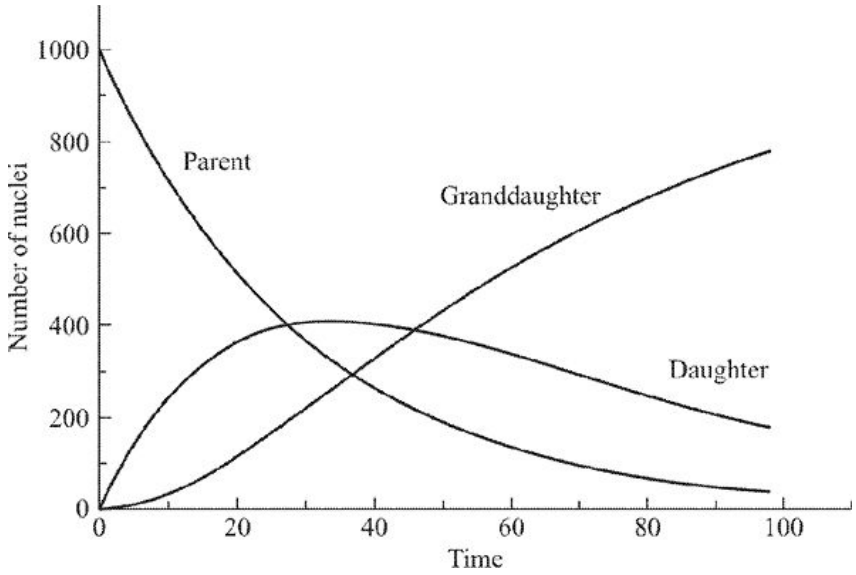


Figure 3.7 Decay of parent, decay and growth of daughter and growth of granddaughter nuclei.

3.4.1 Radioactive Equilibrium

Using Eqs. (3.21), (3.22) and (3.23), one can define two types of equilibria depending upon the decay constants. First case is when $l_1 << l_2$, i.e. when decay constant of the parent nucleus is very small compared to the daughter nucleus. Second is when $l_1 < l_2$. Other possible cases like $l_1 \square l_2$, $l_1 > l_2$ and $l_1 >> l_2$ are also there. The first two cases lead to secular and transient equilibrium respectively. While the other possible cases do not approach equilibrium. We discuss these cases in the following sections.

Permanent or Secular Equilibrium

For $l_1 << l_2$ or $t_1 >> t_2$, where t_1 and t_2 are half-lives of parent and daughter nuclei.

$$\lambda_2 - \lambda_1 \approx \lambda_2$$

and

$$e^{-\lambda_1 t} \approx 1$$

Using these values, Eq. (3.22) becomes

$$N_2 = \frac{\lambda_1}{\lambda_2} N_1^0 (1 - e^{-\lambda_2 t}) \tag{3.24}$$

If t is large compared to half-life of the daughter, i.e. $t >> t_2$, the exponential factor in Eq. (3.24) becomes negligible compared to 1. In this case Eq. (3.24) reduces to

$$N_2 = \frac{\lambda_1}{\lambda_2} N_1^0 \tag{3.25}$$

Since the half-life of the parent nuclei is much larger than the time t, so $N_1{}^0$ almost remains constant, i.e. $N_1{}^0 = N_1$, therefore, Eq. (3.25) becomes

$$\boxed{N_2 \lambda_2 = N_1 \lambda_1 = \text{constant}} \tag{3.26}$$

From this equation it is clear that the amount of daughter N_2 is always constant because N_1, l_1 and l_2 are constants. Since $\frac{N_2}{N_1}$ is constant, the daughter is said to be in permanent or secular or long-term equilibrium with the parent. Equation (3.26) can also be written in the following form using Eq. (3.8)

$$\boxed{\frac{N_2}{N_1} = \frac{\lambda_1}{\lambda_2} = \frac{t_2}{t_1}} \tag{3.27}$$

This equation shows that the number of parent nuclei is inversely proportional to its decay constant or directly proportional to its half-life. Similar argument is true for daughter nucleus. Similarly, if we consider that the daughter is unstable, and it decays into granddaughter, then it can be shown that

$$\boxed{N_1\lambda_1 = N_2\lambda_2 = N_3\lambda_3 = \text{constant}} \tag{3.28}$$

From this equation it is clear that the product of number of parent nuclei and its decay constant is always constant. Some examples of secular equilibrium are

$$^{90}_{38}\text{Sr} \xrightarrow{\beta^-} {}^{90}_{39}\text{Y} \xrightarrow{\beta^-} {}^{90}_{40}\text{Zr}$$

$$^{226}_{88}\text{Ra} \xrightarrow{\alpha} {}^{222}_{86}\text{Rn} \xrightarrow{\alpha} {}^{218}_{84}\text{Po}$$

In Figure 3.8 we have plotted time versus number of atoms of the parent, daughter and granddaughter nuclei for ^{90}Sr decay. In Figure 3.9 we have plotted the activities of parent, daughter and total activity, i.e. parent and daughter activity versus time.

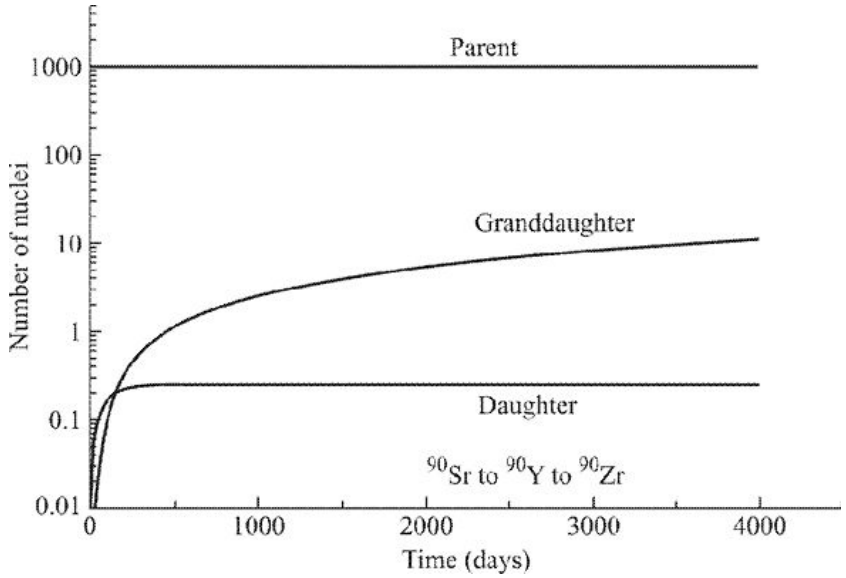


Figure 3.8 A plot of time versus number of atoms of the parent, daughter and granddaughter for ^{90}Sr decay.

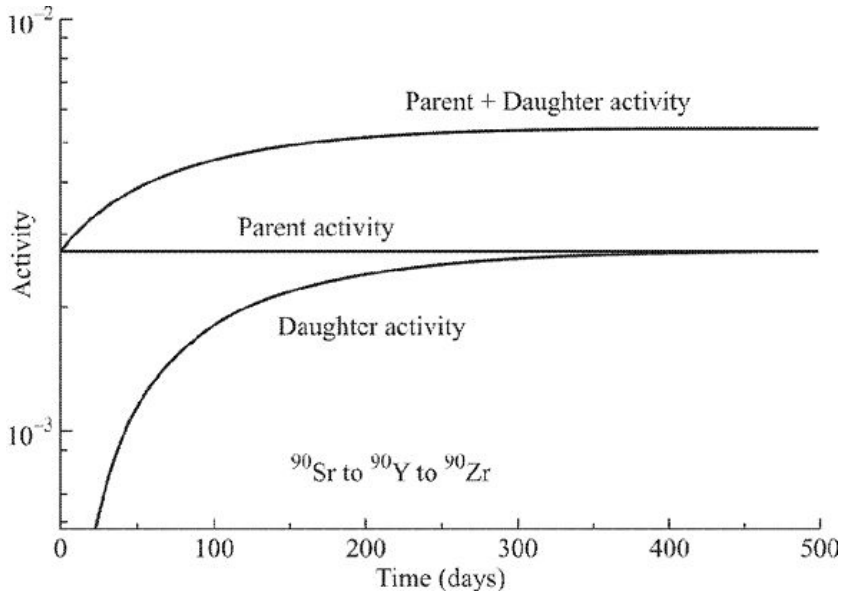


Figure 3.9 A plot of activities of parent, daughter and parent + daughter versus time for ^{90}Sr decay.

Transient Equilibrium

In case $l_1 < l_2$ or $t_1 > t_2$, the number of daughter nuclei must be calculated using Eq. (3.22) for small t. For this case where t becomes sufficiently large, the factor $e^{-l_2 t}$ becomes very small compared to $e^{-l_1 t}$. After this time a state of equilibrium known as transient equilibrium is reached. Neglecting $e^{-l_2 t}$ in comparison to $e^{-l_1 t}$ Eq. (3.22) becomes

$$N_2 = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1^0 e^{-\lambda_1 t} \tag{3.29}$$

Rearranging Eq. (3.29) becomes

$$N_2 = N_1 \frac{\lambda_1}{\lambda_2 - \lambda_1}$$

or

$$\boxed{\frac{N_2}{N_1} = \frac{\lambda_1}{\lambda_2 - \lambda_1} = \text{constant}} \tag{3.30}$$

Some examples of this case are:

$$^{140}\text{Ba} \xrightarrow{\beta^-} {}^{140}\text{La} \xrightarrow{\beta^-} {}^{140}\text{Ce}$$

$$^{132}\text{Te} \xrightarrow{\beta^-} {}^{132}\text{I} \xrightarrow{\beta^-} {}^{132}\text{Xe}$$

In Figure 3.10 we have plotted time versus number of atoms of the parent, daughter and granddaughter nuclei for ^{140}Ba decay. In Figure 3.11 we have plotted the activities of parent, daughter and total activity, i.e. parent and daughter activity versus time.

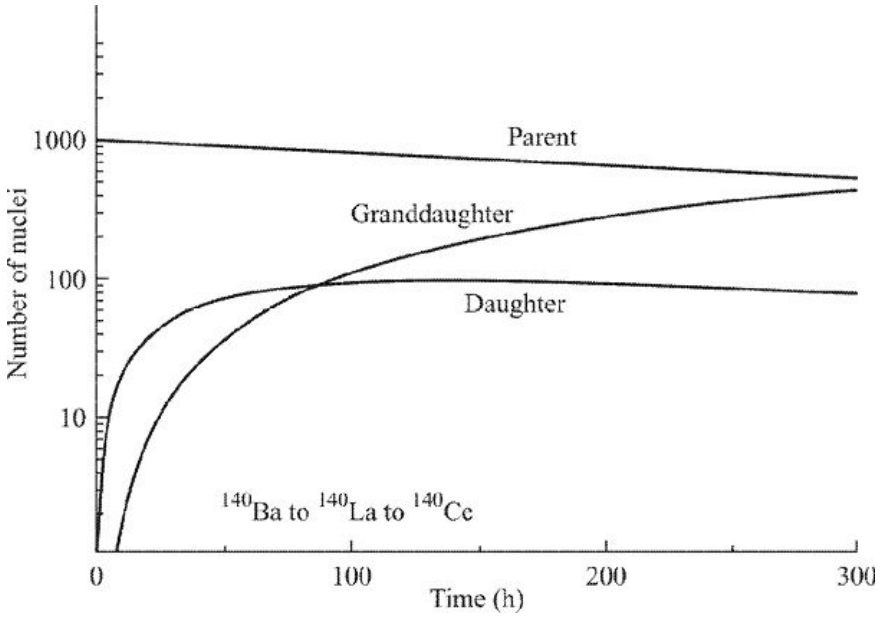


Figure 3.10 Transient equilibrium for ^{140}Ba to ^{140}La. Initially, we start with ^{140}Ba.

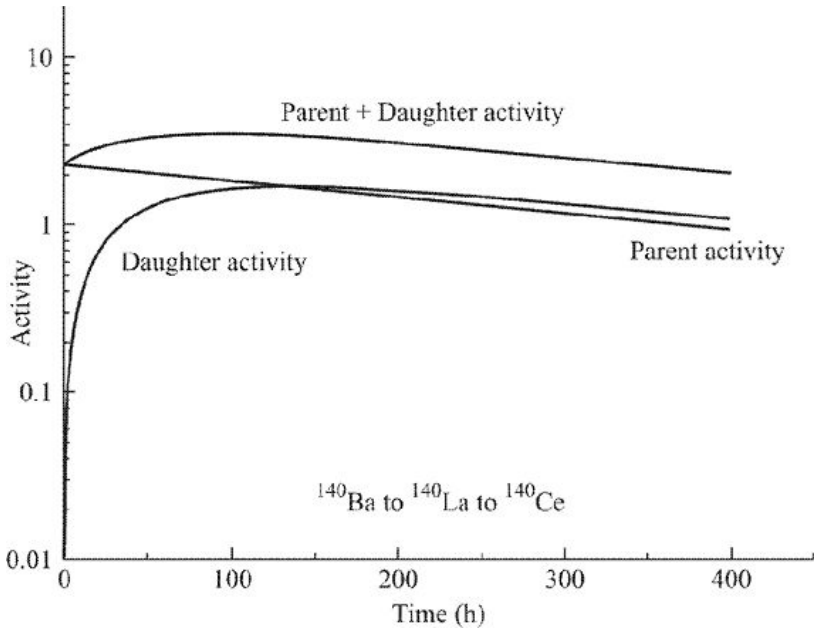


Figure 3.11 A plot of activities of parent, daughter and parent + daughter versus time for ^{140}Ba decay.

Nonequilibrium Case

In case the above conditions of secular and transient equilibria are not met, then we do not approach equilibrium. Some examples of these cases are

$$^{210}_{83}\text{Bi} \xrightarrow{\beta^-} {}^{210}_{84}\text{Po} \xrightarrow{\beta^-} {}^{210}_{85}\text{At}$$

This decay is shown in Figure 3.12.

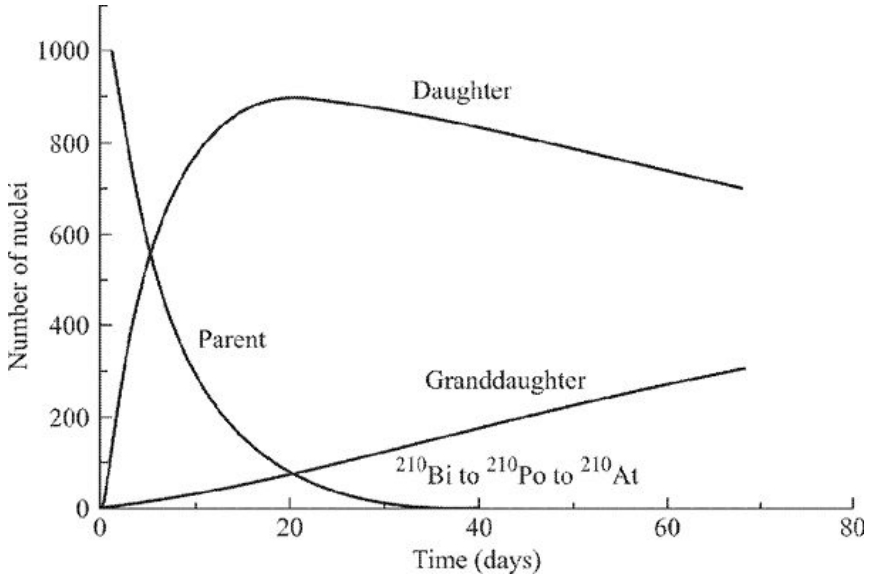


Figure 3.12 A plot of number of nuclei of parent, daughter and granddaughter versus time depicting nonequilibrium for ^{210}Bi decay.

3.5 ALPHA EMISSION

Alpha emission is a process in which parent nucleus spontaneously disintegrates into a daughter nucleus and an *a*-particle.

Rutherford showed that *a*-particles are $^4_2$ He nuclei. Therefore, in this process parent nucleus loses 4 nucleons, its mass decreases by 4 amu and charge decreases by 2. The equation for this decay is

$$\boxed{{}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\text{He} + Q} \qquad (3.31)$$

where Q is the total energy released in this decay process and is called the *disintegration energy*. In order to evaluate Q let us assume that M_p is the mass of parent nucleus M_d is the mass of daughter nucleus and *ma* is the mass of *a*-particle. The parent nucleus is at rest before the emission of *a*-particle, i.e. its kinetic energy is zero. When *a*-particle is emitted with

velocity *va*, to conserve momentum, the daughter nucleus must recoil with velocity V_d. Therefore, by applying law of conservation of momentum

Initial momentum = Final momentum

$0 = M_d V_d - mava$

Negative sign is taken as daughter nucleus and *a*-particles move in opposite directions.

or

$$V_d = \frac{m_\alpha v_\alpha}{M_d}$$

Energy evolved or *Q*-value of the decay is defined as

$$Q = \text{Final kinetic energy} - \text{Initial kinetic energy}$$

$$= \left(\frac{1}{2} M_d V_d^2 + \frac{1}{2} m_\alpha v_\alpha^2\right) - 0$$

Therefore,

$$Q = \frac{1}{2} M_d V_d^2 + \frac{1}{2} m_\alpha v_\alpha^2$$

Substituting for V_d

$$Q = \frac{1}{2} M_d \left(\frac{m_\alpha v_\alpha}{M_d}\right)^2 + \frac{1}{2} m_\alpha v_\alpha^2$$

or

$$= \frac{1}{2} m_\alpha v_\alpha^2 \left(\frac{m_\alpha}{M_d} + 1\right)$$

If *Ka* is the kinetic neergy of *a*-particles or

$$K_\alpha = \frac{1}{2} m_\alpha v_\alpha^2$$

Therefore,

$$Q = K_\alpha \left(\frac{m_\alpha}{M_d} + 1\right)$$

$$\frac{m_\alpha}{M_d} \approx \frac{4}{A-4}$$

…

Therefore,

$$Q = K_\alpha \frac{A}{A-4}$$

or

$$\boxed{K_\alpha = \frac{Q(A-4)}{A}} \tag{3.32}$$

Some common examples of *a*-emitters are:

$$^{241}_{94}\text{Am} \rightarrow ^{237}_{92}\text{U} + ^{4}_{2}\text{He}$$

$$^{226}_{88}\text{Ra} \rightarrow ^{222}_{86}\text{Rn} + ^{4}_{2}\text{He}$$

3.5.1 Properties of Alpha Particles

1. Alpha particles are $^4_2$He nuclei with 2 units of positive charge and mass of 4 amu.
2. Because they are charged particles so they can be deflected by electric and magnetic fields.
3. They ionize the medium through which they pass. Their ionizing power is much higher than *b*- and *g*-rays.
4. Because of the high-ionizing power, they can be easily absorbed by few centimetres of air or fraction of millimetre thick aluminium.
5. Naturally occurring *a*-emitters emit *a*-particles with energy in the range of 5 MeV to
   10 MeV.
6. Alpha particles when fall on certain materials like ZnS, barium platinocynide, etc. emit flashes of light called scintillations. This property of scintillation was initially used to detect *a*-particles.
7. Long exposures to *a*-emitters produce harmful effects on the human body.

3.5.2 Alpha Spectrum

There are only two products in the decay of parent nucleus—daughter nucleus and *a*-particle. Therefore, energy evolved in *a*-decay or *Q*-value is shared by these two particles in the inverse proportion to their masses. Thus,

a-particle carries more energy as compared to the daughter nucleus. These values of the energy of *a*-particle and daughter nucleus are unique. As an example, suppose disintegration energy in the decay of a nucleus is 5.085 MeV, the energy carried by the *a*-particle is 5.000 MeV, while the daughter nucleus carries 0.085 MeV energy. All the *a*-particles emitted in this decay have the same energy. If we plot a curve between the energy of *a*-particle and their number, it would be a straight line as shown in Figure 3.13. This curve is known as *energy spectrum* or *simply spectrum*.

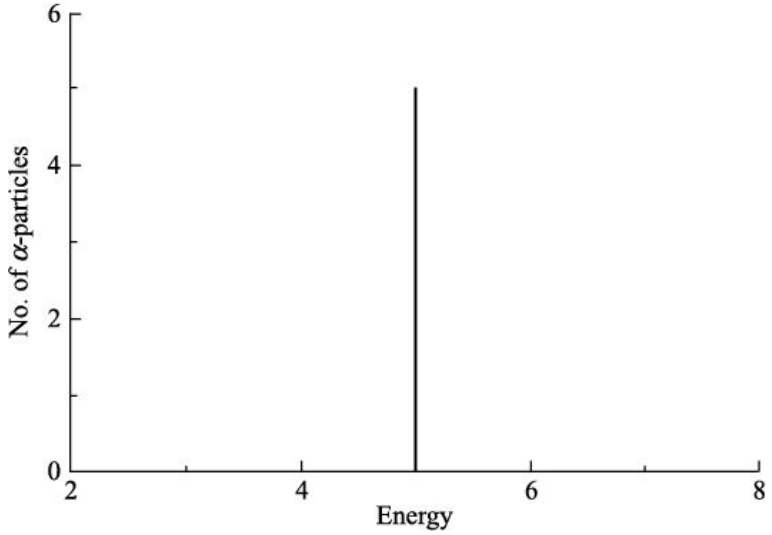


Figure 3.13 Alpha spectrum, a plot between number of *a*-particles versus energy of *a*-particles as recorded with an ideal detector.

The energy spectrum of *a*-particles is also known as line spectrum or discrete spectrum. In actual experiment instead of straight line we observe a broad peak with a maxima at
5.000 MeV (in the example taken above). This broadening is explained on the fact that detection of *a*-particles in a detector is a statistical process. This means that there is a finite probability that all the *a*-particles may not produce same number of electron–ion pairs in the detector medium. It is due to this variation in the number of electron–ion pair produced that the straight line gets broadened.

The actual experimental *a*-particle spectrum in the decay of ^{241}Am as recorded by a semi-conductor detector is shown in Figure 3.14. In this spectrum, there are two peaks, first at
5.48 MeV and second at 5.44 MeV. These two peaks correspond to two

groups of *a*-particles emitted by $^{241}_{95}$ Am source. The *a*-decay of ^{241}Am is represented as

$$^{241}_{95}\text{Am} \rightarrow {}^{237}_{93}\text{Np} + {}^{4}_{2}\text{He}$$

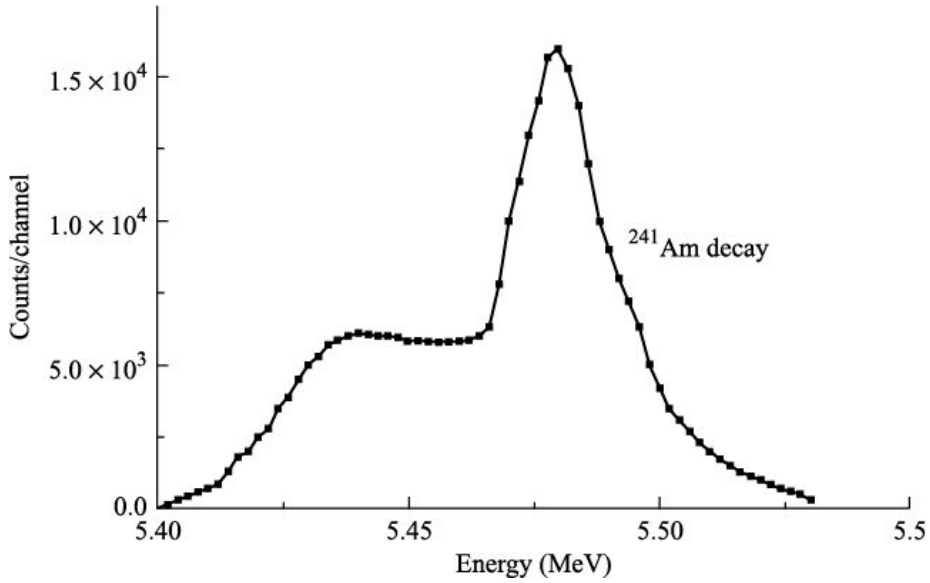


Figure 3.14 Actual alpha spectrum as observed in the decay of ^{241}Am. The two peaks at 5.44 and 5.48 MeV correspond to two groups of *a*-particles emitted by ^{241}Am.

When 5.48 MeV *a*-particle is emitted, ^{237}Np is left in the first excited state having energy 0.0595 MeV. When 5.44 MeV *a*-particle is emitted, ^{237}Np is left in the second excited state having energy 0.103 MeV. The second excited state of ^{237}Np can make a direct transition to the ground state by emitting a *g*-ray of energy 0.103 MeV, or to the first excited state by emitting a *g*-ray of energy 0.0434 MeV. The first excited state makes a transition to the ground state by emitting a *g*-ray of energy 0.0595 MeV. In most *a*-emitting nuclei, many groups of *a*-particles are emitted. A diagram showing the ground state of parent *a*-emitting nucleus and different excited states of daughter nucleus populated by *a*-particles is known as decay scheme of the nucleus. The decay scheme of ^{241}Am is shown in Figure 3.15.

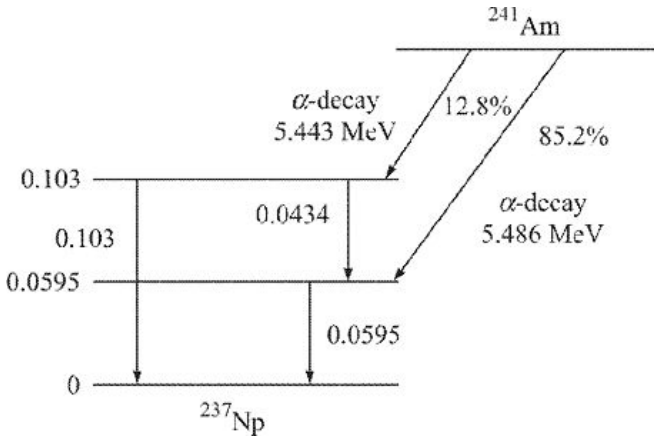


Figure 3.15 Decay scheme of ^{241}Am.

3.5.3 Range of Alpha Particles

If *a*-particles emitted by a radioactive source are counted by counting the number of scintillations produced on ZnS screen, it is observed that the number of *a*-particles hitting the screen are practically constant up to a certain distance *R* from the source and then suddenly drop to zero. This distance *R* is called the range of *a*-particles.

Range is defined as the distance moved by an a-particle in a given material before it comes to rest.

It is related to the initial energy of *a*-particles. Greater the initial energy of *a*-particles, greater is the range and vice versa. If instead of air, we take some other material like aluminium or iron, etc. the range *R* is different. Empirically, it has been found that the range of *a*-particles in a medium is inversely proportional to the density of the medium and directly proportional to , where *A* is the atomic weight of the medium.

3.5.4 Geiger–Nuttal Law

Geiger and Nuttal in 1911 noticed that *a*-emitters with large disintegration energy have short half-lives and vice versa. They observed that a change of factor 2 in energy scale changes the half-life by a factor of 10^{24}. As an example ^{232}Th has a half-life of 1.4 □ 10^{10} years (4.4 □ 10^{17} s) and energy of *a*-particles emitted by it is 4.08 MeV. ^{218}Th has a half-life of 10^{-7} s while energy of the emitted *a*-particle is 9.85 MeV. In other words, disintegration constant varies extremely rapidly with small changes in energy. The longest-lived *a*-emitters emit least energetic *a*-particles and vice versa.

These empirical facts were correlated by Geiger–Nuttal in 1911 when log(l) was plotted against log(R). Plots for members of the three naturally occurring radioactive series (discussed in Section 3.3) were found to fall along three parallel straight lines, one line for each series as shown in Figure 3.16. These straight lines are expressed by Geiger–Nuttal law which is as under

$$\log(\lambda) = A \times \log(R) + B \tag{3.33}$$

where A and B are constants. Value of A is nearly the same for all the three lines but B has different values.

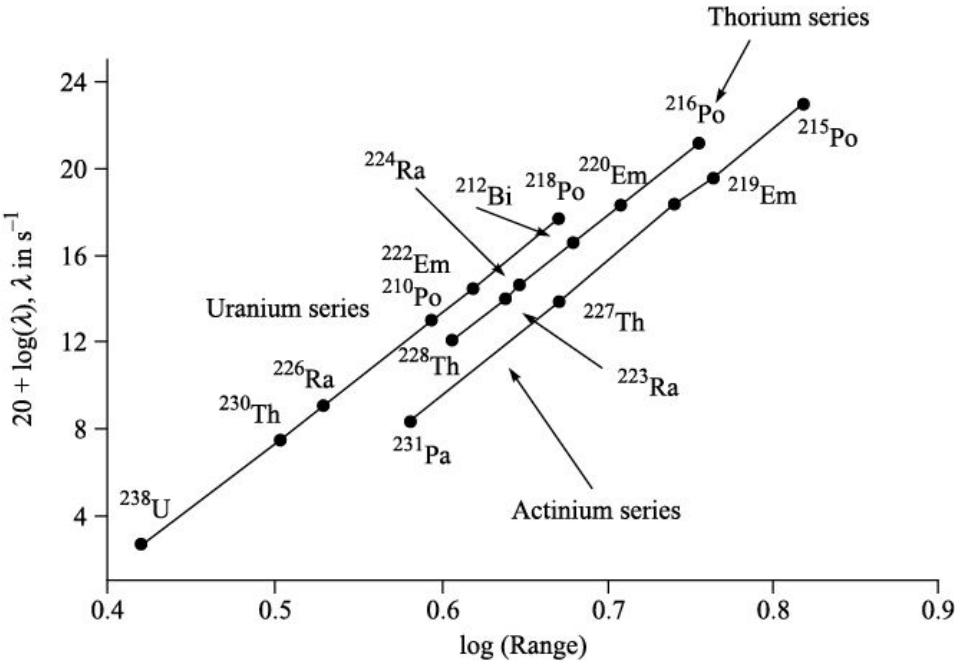


Figure 3.16 Plot of log(l) versus log(Range) for uranium, thorium and actinium series.

In terms of energy, Geiger–Nuttal law is

$$\log(\lambda) = A' \times \log(E) + B' \tag{3.34}$$

The constants A and B of Eq. (3.33) become A□ and B□ when the range is converted to energy.

3.5.5 Gamow Theory of Alpha Decay

Let us discuss the scattering of a-particles from a nucleus. In scattering experiments as a-particle from infinity comes in close proximity of scattering

nucleus, the Coulomb repulsion between the two increases. When the nucleus and *a*-particle are just touching each other, the repulsion is maximum. If the kinetic energy of *a*-particle is larger than the Coulomb repulsion, the *a*-particle will overcome the repulsion and will enter the nucleus. At this point, suddenly the *a*-particle enters the region of nuclear force, where it is attracted by the nucleons in the nucleus. This interaction between the nucleus and an *a*-particle can be represented pictorially by a potential energy curve as shown in Figure 3.17. In this figure, the rising portion from *A* to *B* indicates increasing electrostatic repulsion of an *a*-particle as it approaches the nucleus. The sharp fall from *B* to *C* is essentially the region within the nucleus, implies attraction of the *a*-particle by the nucleons present in the nucleus.

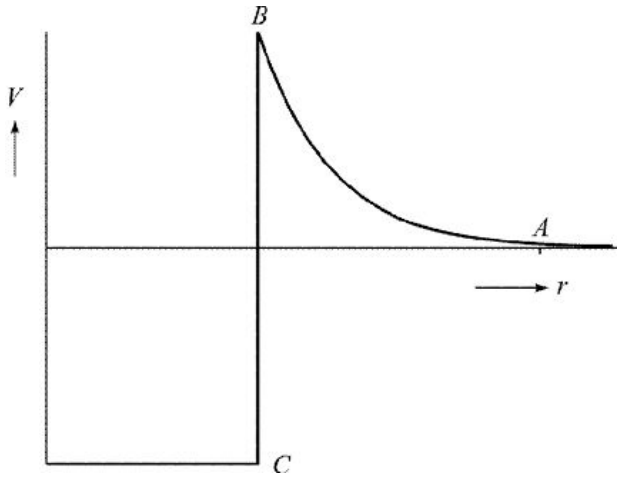


Figure 3.17 Interaction between *a*-particle and the nucleus represented by potential energy curve.

Calculations based on Coulomb law show that the point *B* corresponds to energy of about
25 MeV for higher *Z* elements. Hence, it may be concluded that an *a*-particle of energy less than 25 MeV should be repelled if it approaches the nucleus from outside. If this potential barrier prevents the access of *a*-particle from outside the nucleus, the same barrier should prevent the emission of *a*-particles from the interior of the nucleus. However, it is surprising that some radioactive elements emit *a*-particles with energy as low as 4.0 MeV, and for no natural radioisotope the *a*-particle energy exceeds 11.0 MeV.

Classical mechanics has no explanation to *a*-decay process. In 1928, Gurney and Condon of U.S.A., and Gamow of Russia independently

developed a theory known as Gamow theory of *a*-decay to tackle the problem of *a*-decay using quantum mechanics.

According to classical mechanics an *a*-particle having energy 5–10 MeV inside the nucleus cannot cross the potential barrier and escape, if its energy is less than the maximum height of the Coulomb barrier which is about 25 MeV, but quantum mechanics predicts that there is a definite but small probability that such a particle from the interior can be found outside the nucleus. In other words, there is a definite probability that the *a*-particle can escape from the nucleus even when its energy is less than that of the top of the potential barrier. Thus, quantum mechanically, there is finite probability that *a*-particle leaks through the potential barrier, as if there are holes in the potential barrier. This effect is called *quantum mechanical tunnelling* or *simply tunnelling*.

In order to explain the phenomenon of *a*-decay, Gamow made the following assumptions:

- An *a*-particle may exist as an *a*-particle within a heavy nucleus.
- Such an *a*-particle is in constant motion and is held in the nucleus by a potential barrier.
- There is a small but definite likelihood that the *a*-particle may tunnel through the barrier (despite the height of the barrier) each time it collides with the barrier.

Based on these assumptions, Gamow derived the relation for the probability that *a*-particle may tunnel through the potential barrier as under.

In order to make the calculations simple, we consider only one-dimensional rectangular potential barrier of width a and height V, which is greater than the kinetic energy of *a*-particle. This rectangular potential barrier is shown in Figure 3.18. There are three regions of interest marked as I, II, and III. In region I, incident and reflected *a*-particles or waves associated with *a*-particles are present. In region II, both forward moving transmitted waves and reflected waves from the other side of the barrier are present. In region III, only forward moving transmitted waves are present. Mathematically, the rectangular potential is defined as

$$V(x) = 0 \qquad \text{for } x < 0$$
$$V(x) = V \qquad \text{for } 0 \leq x \leq a$$

$V(x) = 0 \qquad \text{for } x > a$

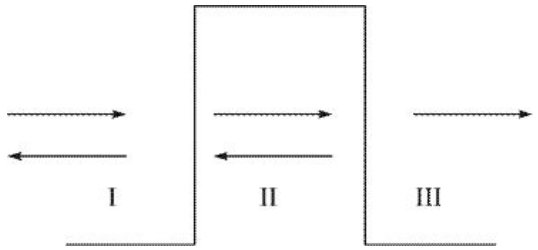


Figure 3.18 The rectangular potential barrier $V(x)$ versus x.

One dimensional time independent Schrödinger equation is given as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

In regions I and III, $V = 0$, so one-dimensional time-independent Schrödinger equation becomes

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0 \qquad (3.35)$$

where $m = \frac{m_a m_d}{m_a + m_d}$ is the reduced mass of the a-particle and daughter nucleus; $\hbar = \frac{h}{2p}, h$ being the Planck constant. As both incident and reflected a-waves are present in region I, so the solution of Eq. (3.35) in region I is

$$y_I = Ae^{ik1x} + Be^{-ik1x} \qquad (3.36)$$

where

$$k_1 = \frac{\sqrt{2mE}}{\hbar} = \frac{p}{\hbar} = \frac{2\pi}{\lambda} \qquad (3.37)$$

Here p is the momentum and l is the de Broglie wavelength associated with a-particles of energy E. A and B are constants to be evaluated later on. The first term in the solution (Eq. 3.36) represents the incident wave and the second term represents the reflected wave.

In region III, only the forward moving transmitted wave is present, hence the solution of
Eq. (3.35) is

$$y_{III} = Ce^{ik1x} \qquad (3.38)$$

Here C is constant.

The one-dimensional time-independent Schrödinger equation for the region II, where $E < V$ is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

or

$$\frac{d^2\psi}{dx^2} - \frac{2m}{\hbar^2}(V - E)\psi = 0 \qquad (3.39)$$

The solution of this equation in region II is

$$\psi_{II} = Fe^{-k_2x} + Ge^{k_2x} \qquad (3.40)$$

First term represents reflected wave and second term represents refracted wave. Here F and G are constants to be determined and

$$k_2 = \frac{\sqrt{2m(V - E)}}{\hbar} \qquad (3.41)$$

In order to evaluate constants A, B, C, F and G, we apply certain conditions known as boundary conditions. These conditions are:

1. Wave functions of the two regions must be same or match at the boundary of the two regions, i.e. at $x = 0$ and at $x = a$.
2. First derivative of wave functions of the two regions must be same or match at the boundary of the two regions, i.e. at $x = 0$ and at $x = a$.

Combining these two conditions, we get a new set of conditions i.e.,

Condition 1

$$\psi_I = \psi_{II} \text{ at } x = 0 \quad \text{and} \quad \frac{d\psi_I}{dx} = \frac{d\psi_{II}}{dx} \text{ at } x = 0$$

Condition 2

$$\psi_{II} = \psi_{III} \text{ at } x = a \quad \text{and} \quad \frac{d\psi_{II}}{dx} = \frac{d\psi_{III}}{dx} \text{ at } x = a$$

Applying condition 1 to Eqs. (3.36) and (3.40), we get

$$A + B = F + G \tag{3.42}$$

and

$$ik_1(A - B) = -k_2(F - G) \tag{3.43}$$

Similarly, applying condition 2 to Eqs. (3.38) and (3.40), we get

$$Ce^{ik_1a} = Fe^{-k_2a} + Ge^{k_2a} \tag{3.44}$$

and

$$ik_1Ce^{ik_1a} = -Fk_2e^{-k_2a} + Gk_2e^{k_2a} \tag{3.45}$$

Dividing Eq. (3.45) by k_2 and adding to Eq. (3.44), we get

$$Ce^{ik_1a}\left(\frac{ik_1}{k_2} + 1\right) = 2Ge^{k_2a}$$

or

$$G = \frac{Ce^{ik_1a}}{2}\left(\frac{ik_1}{k_2} + 1\right)e^{-k_2a} \tag{3.46}$$

Similarly, dividing Eq. (3.45) by k_2 and subtracting from Eq. (3.44), we get

$$F = \frac{Ce^{ik_1a}}{2}\left(1 - \frac{ik_1}{k_2}\right)e^{k_2a} \tag{3.47}$$

Substituting G and F from Eqs. (3.46) and (3.47) in Eq. (3.42), we get

$$A + B = \frac{Ce^{ik_1a}}{2}\left(1 - \frac{ik_1}{k_2}\right)e^{k_2a} + \frac{Ce^{ik_1a}}{2}\left(\frac{ik_1}{k_2} + 1\right)e^{-k_2a}$$

Rearranging this equation, we get

$$A + B = \frac{C}{2}e^{ik_1a}(e^{-k_2a} + e^{k_2a}) + \frac{C}{2}\frac{ik_1}{k_2}e^{ik_1a}(e^{-k_2a} - e^{k_2a}) \tag{3.48}$$

We have

$$2\cos hx = e^x + e^{-x} \text{ and } 2\sin hx = e^x - e^{-x}$$

Therefore,

$$A + B = Ce^{ik_1a}\left(\cos hk_2a - \frac{ik_1}{k_2}\sin hk_2a\right) \tag{3.49}$$

Similarly, substituting G and F from Eqs. (3.46) and (3.47) in Eq. (3.43), we get

$$A - B = \frac{-k_2}{ik_1}\frac{C}{2}e^{ik_1a}\left[(e^{k_2a} - e^{-k_2a}) - \frac{ik_1}{k_2}(e^{k_2a} + e^{-k_2a})\right]$$

or

$$A - B = \frac{-k_2}{ik_1}Ce^{ik_1a}\left[\frac{e^{k_2a} - e^{-k_2a}}{2} - \frac{ik_1}{k_2}\frac{e^{k_2a} + e^{-k_2a}}{2}\right]$$

or

$$A - B = \frac{-k_2}{ik_1}Ce^{ik_1a}\left(\sin h(k_2a) - \frac{ik_1}{k_2}\cos h(k_2a)\right)$$

or

$$A - B = Ce^{ik_1a}\left(\cos h(k_2a) + \frac{ik_2}{k_1}\sin h(k_2a)\right) \tag{3.50}$$

Adding Eqs. (3.49) and (3.50), we get

$$\frac{A}{C} = e^{ik_1a}\left[\cos h(k_2a) + \frac{i}{2}\left(\frac{k_2}{k_1} - \frac{k_1}{k_2}\right)\sin h(k_2a)\right] \tag{3.51}$$

Subtracting Eqs. (3.49) and (3.50), we get

$$\frac{B}{C} = -\frac{i}{2}e^{ik_1a}\left[\frac{k_1}{k_2} + \frac{k_2}{k_1}\right]\sin h(k_2a) \tag{3.52}$$

Let v_1 be the velocity of incident flux of *a*-particles in region I and v_3 the velocity of transmitted flux of *a*-particles in region III. Since regions I and III are force free, so $v_3 = v_1$. Transmission coefficient T is given by

$$T = \frac{CC^*}{AA^*}$$

$$T = \frac{1}{\cos h(k_2a) + \frac{i}{2}\left(\frac{k_2}{k_1} - \frac{k_1}{k_2}\right)\sin h(k_2a)} \times \frac{1}{\cos h(k_2a) - \frac{i}{2}\left(\frac{k_2}{k_1} - \frac{k_1}{k_2}\right)\sin h(k_2a)}$$

or

$$T = \frac{1}{\cos h^2(k_2a) + \frac{1}{4}\left(\frac{k_2}{k_1} - \frac{k_1}{k_2}\right)^2 \sin h^2(k_2a)} \tag{3.53}$$

Now,

$$\cos h^2(x) - \sin h^2(x) = 1$$

or $\cosh^2(x) = 1 + \sinh^2(x)$

Therefore,

$$T = \frac{1}{1 + \frac{1}{4}\frac{(k_2^2 + k_1^2)^2}{k_1^2 k_2^2}\sinh^2(k_2 a)} \tag{3.54}$$

From Eqs. (3.37) and (3.41), we have

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

$$k_2 = \frac{\sqrt{2m(V-E)}}{\hbar}$$

$$k_1^2 + k_2^2 = \frac{2mV}{\hbar^2} \quad \text{and} \quad k_1 k_2 = \frac{2m}{\hbar^2}\sqrt{E(V-E)}$$

Therefore, Eq. (3.54) becomes

$$T = \frac{1}{1 + \frac{1}{4}\frac{V^2}{E(V-E)}\sinh^2(k_2 a)}$$

$$T = \left[1 + \frac{1}{4}\frac{V^2}{E(V-E)}\sinh^2(k_2 a)\right]^{-1} \tag{3.55}$$

Since $k_2 a >> 1$ and $\sinh^2(x) = \frac{1}{4}(e^{2x} + e^{-2x} - 2)$

So, $\sinh^2(k_2 a) = \frac{1}{4}e^{2k_2 a}$, where we have neglected $e^{-2k_2 a} - 2$, since this term is much smaller than $e^{2k_2 a}$. Thus, Eq. (3.55) becomes

$$T = \left[1 + \frac{1}{16}\frac{V^2}{E(V-E)}e^{2k_2 a}\right]^{-1}$$

Now, $V > E$ and $e^{2k_2 a} >> 1$, so we can neglect 1 in comparison to the second term of the above equation. So, we have

$$T = \left[\frac{1}{16}\frac{V^2}{E(V-E)}e^{2k_2 a}\right]^{-1} = 16\frac{E(V-E)}{V^2}e^{-2k_2 a} \tag{3.56}$$

When $2k_2a >> 1$ the exponential factor in the above equation becomes extremely small, while the factor before the exponential is of the order of 1. For the order of magnitude calculations

$$T = e^{-2k_2a} \tag{3.57}$$

or

$$\log_e(T) = -2k_2a \tag{3.58}$$

Equation (3.57) represents the fraction of *a*-particles that will penetrate the barrier of width *a* and of height $V > E$. Inside the nucleus the *a*-particle is faced with a barrier of varying height. Because of the varying height Eq. (3.58) can be expressed as an integral, i.e.

$$\log_e(T) = -2\int_{r_0}^{r} k_2 x dx$$

where r_0 is the distance from the centre of the nucleus for which potential is constant. r is the distance from the centre of the nucleus where the potential energy becomes equal to E. For
$x > r$, $E > V$, i.e. *a*-particle can escape permanently from the nucleus.
To calculate the integral, we define the potential energy $V(x)$ as

$$V(x) = \frac{2(Z-2)e^2}{4\pi\varepsilon_0 x}$$

where $(Z - 2)e$ is the charge of the daughter nucleus and $2e$ is the charge of *a*-particle.
Now,

$$k_2 = \frac{\sqrt{2m(V-E)}}{\hbar} = \left(\frac{2m}{\hbar^2}\right)^{1/2}\left(\frac{2(Z-2)e^2}{4\pi\varepsilon_0 x} - E\right)^{1/2}$$

Since $V = E$ at $x = r$, so

$$E = \frac{2(Z-2)e^2}{4\pi\varepsilon_0 r} \tag{3.59}$$

Thus,

$$k_2 = \left(\frac{2mE}{\hbar^2}\right)^{1/2}\left(\frac{r}{x} - 1\right)^{1/2}$$

Now,

$$\log_e(T) = -2\int_{r_0}^{r}\left(\frac{2mE}{\hbar^2}\right)^{1/2}\left(\frac{r}{x} - 1\right)^{1/2} dx$$

Solving this integral, we get

$$\log_e(T) = -2\left(\frac{2mE}{\hbar^2}\right)^{1/2} r\left\{\cos^{-1}\left(\frac{r_0}{r}\right)^{1/2} - \left(\frac{r_0}{r}\right)^{1/2}\left(1 - \frac{r_0}{r}\right)^{1/2}\right\}$$

The potential barrier is relatively wide, i.e. $r >> r_0$. For this approximation, we can write

$$\cos^{-1}\left(\frac{r_0}{r}\right)^{1/2} = \frac{\pi}{2} - \left(\frac{r_0}{r}\right)^{1/2}$$

and

$$\left(1 - \frac{r_0}{r}\right)^{1/2} = 1$$

Thus,

$$\log_e(T) = -2\left(\frac{2mE}{\hbar^2}\right)^{1/2} r\left\{\frac{\pi}{2} - 2\left(\frac{r_0}{2}\right)^{1/2}\right\} \tag{3.60}$$

We have from Eq. (3.59)

$$r = \frac{2(Z-2)e^2}{4\pi\varepsilon_0 E} \tag{3.61}$$

Substituting r from Eq. (3.61) in Eq. (3.60), we get

$$\log_e(T) = \frac{4e}{\hbar}\left(\frac{m}{\pi\varepsilon_0}\right)^{1/2}(Z-2)^{1/2} r_0^{1/2} - \frac{e^2}{\hbar\varepsilon_0}\left(\frac{m}{2}\right)^{1/2}(Z-2)E^{-1/2} \tag{3.62}$$

Substituting the values of various constants, we get

$$\log_e(T) = 2.97(Z-2)^{1/2} r_0^{1/2} - 3.95(Z-2)E^{-1/2} \tag{3.63}$$

where r_0 is expressed in fermi. We also find

Decay constant $l = vT$

or

$$\log_e(l) = \log_e(v) + \log_e(T) \tag{3.64}$$

where n is the number of times the a-particle makes collisions with the potential barrier per unit time. If n is the velocity of a-particle in the nucleus, then the collision frequency is approximately given by

$$v = \frac{v}{2r_0} \tag{3.65}$$

Combining Eqs. (3.63), (3.64) and (3.65), we get

$$\log_e(\lambda) = \log_e\left(\frac{v}{2r_0}\right) + \log_e[2.97(Z-2)^{1/2} r_0^{1/2} - 3.95(Z-2)E^{-1/2}] \tag{3.66}$$

which is the required equation of Gamow theory.

3.6 BETA DECAY

We have already seen (Figure 3.1) one of the modes of radioactive decay is by b-emission. In this mode three types of decay processes are encountered.

1. The nucleus decays by an electron emission. This process is known as b^--emission, e.g. ${}^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + \beta^- + \bar{\nu}$.
2. The nucleus decays by positron emission. This decay is known as b^+-decay, e.g.
   ${}^{22}_{11}\mathrm{Na} \rightarrow {}^{22}_{12}\mathrm{Ne} + b^+ + n$.
3. The nucleus captures one of its orbital electrons. This process is called electron capture, e.g. ${}^{54}_{25}\mathrm{Mn} + b^- \rightarrow {}^{54}_{24}\mathrm{Cr} + n$.

These disintegrations are also called *isobaric transformations* as there is no change in the mass of the nucleus but there is a change of nuclear charge by

one unit.

In Chapter 1, we have seen that nucleus consists of protons and neutrons. Electrons cannot exist inside the nucleus. Therefore, in b^--decay a neutron must be converted into a proton at the time of b^--emission. Thus, in this process the charge of the decaying nucleus increases by 1. In b^+-decay a proton is converted into a neutron. Also in electron capture a proton is converted into a neutron. In both these processes, charge of the nucleus decreases by 1.

3.6.1 Conditions for Spontaneous Emission of b^--Particles

b^--decay is represented by the general equation

$$\boxed{{}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \bar{\nu}}$$

or

$${}^{1}_{0}n \rightarrow {}^{1}_{1}\mathrm{H} + {}^{0}_{-1}e + \bar{\nu}$$

As is well known now, two types of neutrinos exist, one emitted in the decay of nucleon which is denoted as n_e and the second in the decay of muon, which is denoted as nm. For the sake of simplicity, in this discussion of b-decay, we denote the neutrino emitted in nucleon decay
as n.

Let M_X, M_Y and m_e be the rest masses of the parent nucleus, daughter nucleus and electron respectively. Assume that the parent nucleus is at rest. Let K_Y and K_e be the kinetic energies of the daughter nucleus and the electron respectively. Using the law of conservation of energy, we have

$$M_X c^2 = M_Y c^2 + K_Y + m_e c^2 + K_e$$

In this equation, we have neglected the energy carried by neutrino Kn. In b-decay, total energy evolved is shared by nucleon, b^--particle and neutrino in all possible proportions. Here we consider a special case, when neutrino carries zero kinetic energy.
or

$$K_Y + K_e = Q = (M_X - M_Y - m_e)c^2 \qquad (3.67)$$

For spontaneous decay Q must be positive, i.e.

$$M_X > (M_Y + m_e)$$

Thus, for b^--emission rest mass of the parent nucleus must be greater than the sum of rest masses of the daughter nucleus and the electron.
If $M(Z)$ and $M(Z + 1)$ are the rest masses of the parent and daughter atom, then

$$M(Z) = M_X + m_e Z$$
$$M(Z + 1) = M_Y + m_e(Z + 1)$$

In writing these equations, we have neglected the binding energy of the electrons.
Rewriting the above equations in the following form

$$M_X = M(Z) - m_e Z$$
$$M_Y = M(Z + 1) - m_e(Z + 1)$$

Substituting these values in (3.67) equation, we get

$$Q = [M(Z) - M(Z + 1)]c^2 \qquad (3.68)$$

Therefore, b^--decay occurs whenever the mass of the parent atom is greater than that of the daughter atom. In other words, nucleus decays by b^--emission, if it is heavier than another atom with Z greater by one but having same A.

3.6.2 Conditions for Spontaneous Emission of b^+-Particles

b^+-decay is represented by the equation

$$^{A}_{Z}X \square \, ^{A}_{Z-1}Y + ^{0}_{1}e + n$$

or

$$_1{}^1\mathrm{H} \square \, _0{}^1n + _{+1}{}^0e + n$$

As in b^--decay, the disintegration energy Q for b^+-decay is given by the relation

$$Q = K_Y + K_e = (M_X - M_Y - m_e)c^2$$

If $M(Z)$ and $M(Z-1)$ are the rest masses of the parent and daughter atom, then

$$M(Z) = M_X + m_e Z$$
$$M(Z-1) = M_Y + m_e(Z-1)$$

We get

$$Q = [M(Z) - M(Z+1) - 2m_e]c^2 \tag{3.69}$$

For b^+-decay Q must be positive.

Therefore, b^+-decay of atom occurs only if its mass is greater than the sum of the rest mass of the atom with same A but Z less by one and twice the rest mass of an electron.

3.6.3 Beta-Particle Spectrum

In Section 3.6, we have discussed that b-decay is a three body problem, i.e. there are three products in the decay of parent nucleus—daughter nucleus, b-particle and a neutrino. If we look at the energy carried by b-particle, it can carry any amount of energy starting from zero to Q. If b-particle carries zero energy, the remaining energy (Q) will be shared by daughter nucleus and neutrino. If b-particle carries energy Q, then the other two decay products will carry zero energy. If it carries energy between zero and Q, the remaining energy will be shared by the other two decay products. Therefore, the b-particle spectrum is a continuous spectrum unlike a-particle spectrum, which is a line spectrum or discrete spectrum.

Figures 3.19, 3.20 and 3.21 show the b^--, b^+-spectra from $^{64}_{27}$Cu and b^--spectrum from $^{210}_{83}$Bi. All the three figures indicate that the b's emitted in nuclear decay have a continuous energy distribution and energy of b's ranges from zero to certain maximum value called the end point energy. There are certain characteristics of these spectra.

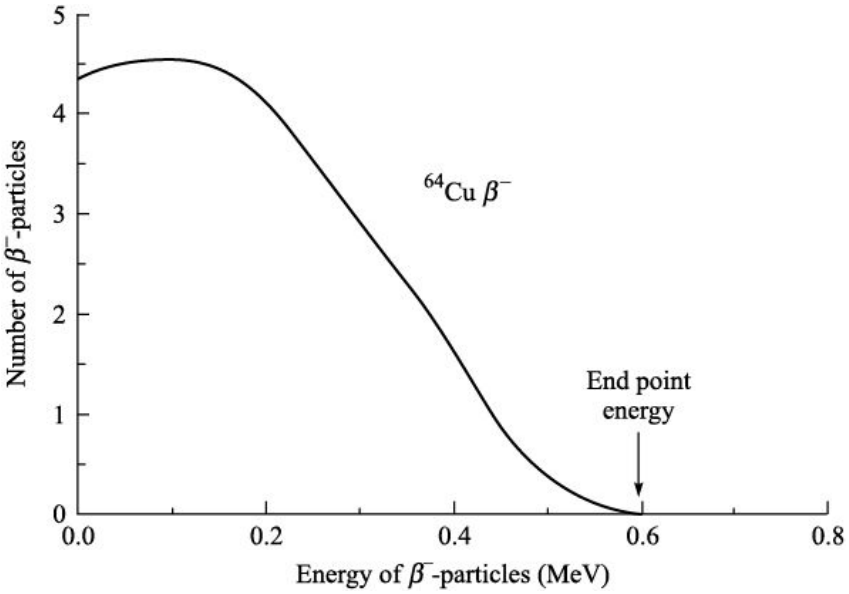


Figure 3.19 Spectrum of b^--particles emitted in the decay of ^{64}Cu.

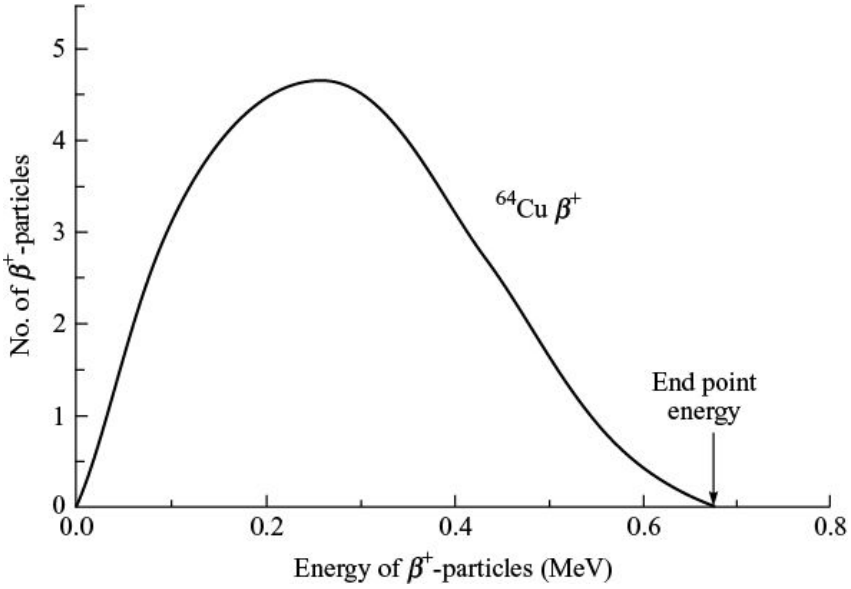


Figure 3.20 Spectrum of b^+-particles emitted in the decay of ^{64}Cu.

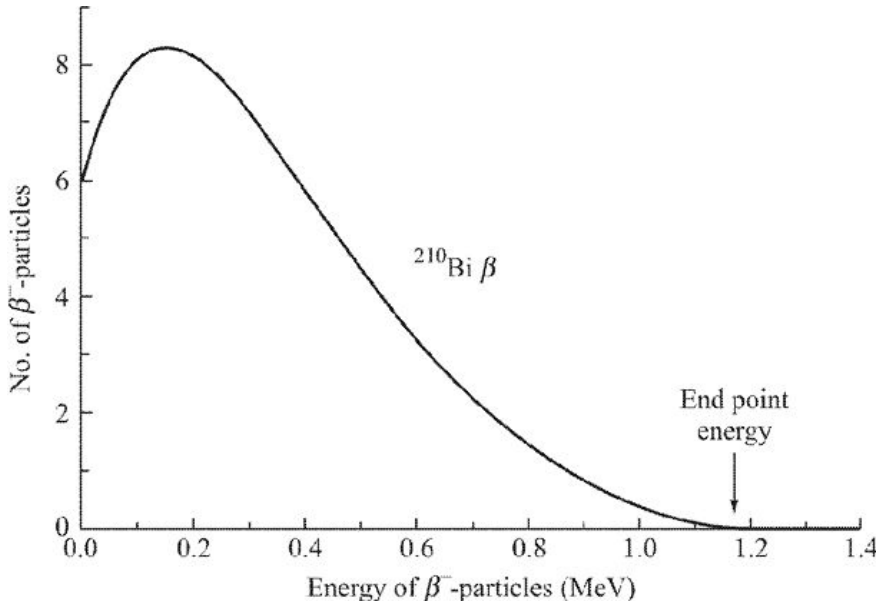


Figure 3.21 Spectrum of b^--particles emitted in the decay of ^{210}Bi.

1. In all these distributions, there is a maximum. The energy of these maxima depends upon the nucleus undergoing *b*-decay.
2. There is definite end point energy. This end point energy depends upon the nucleus undergoing *b*-decay.
3. These continuous *b*-spectra are observed for all natural and man-made *b*-emitters.
4. Figures 3.19 and 3.20 are the spectra of b^-- and b^+-decay. They have been plotted for the same nucleus (^{64}Cu) and have nearly same end point energy. The difference in both the curves is

- That the b^--spectrum is shifted towards to lower energy side.
- That the b^+-spectrum is shifted towards to higher energy side.

This is expected from the fact that the b^--particles are slowed down and b^+-particles are speeded by the Coulomb force of the nucleus. Theoretically, it has been shown that if Coulomb force effect is not taken into account, then both the curves are identical.

Figure 3.22 shows the b^--spectrum of ^{137}Cs. This spectrum again is continuous but there are some sharp peaks or lines superimposed on the continuous distribution. These lines are due to internal conversion, which can be explained as under. In ^{137}Cs b^--decay does not take place to ground state of ^{137}Ba but to an excited state of ^{137}Ba. ^{137}Ba can return to the ground state either by emitting a *g*-ray or it may directly transfer the excess energy to one of the orbital electron, e.g. *K*, *L*, *M,* ..., shall etc. electrons and that electron is ejected from the atom. These electrons are known as conversion electrons. The energy of conversion electron is the total excess energy available with the nucleus minus the binding energy of the electron in the respective shell. So, the conversion electrons have unique energy and hence show in the spectrum as peaks superimposed on the continuous distribution.

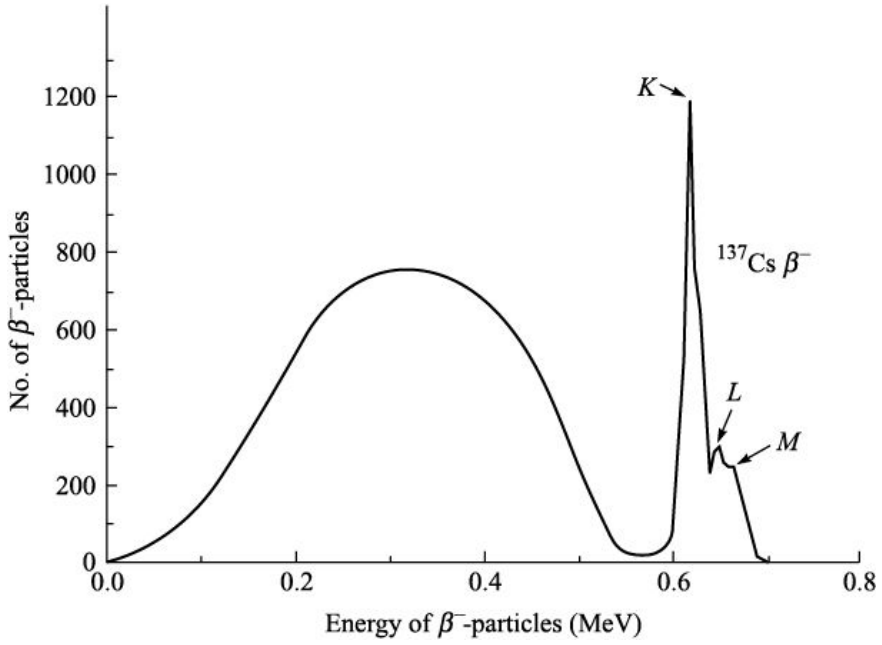


Figure 3.22 Spectrum of b^--particles emitted in the decay of ^{137}Cs.

It is important to mention that atomic number is defined as the number of protons in the nucleus. In internal conversion, the electron is ejected from the electronic orbits, therefore, the atomic number of the nucleus remains the same. This is unlike *b*-decay, where a proton/neutron inside the nucleus decays to neutron/proton.

3.6.4 Electron Capture

This process is represented by the equation

$$ {}^{A}_{Z}X + {}^{0}_{-1}e \square {}^{A}_{Z-1}Y + n $$

Proceeding in the same way as that b^--decay, the disintegration energy for this process is

$$ Q = [M(Z) - M(Z-1)]c^2 \qquad (3.70) $$

Therefore, for electron capture to occur, the mass of the parent atom must be greater than the mass of a nucleus with same A but with Z less by one.

3.6.5 Neutrino (n) and Antineutrino ($\bar{n}$)

As already discussed in b^--decay, during b-decay a new particle neutrino or antineutrino is emitted. Here are some properties of neutrino.[1]

[1] Precisely speaking, these neutrinos are called e-neutrinos or n_e and $\bar{n}_e$. For details see Chapter 8 (section 8.5.1)

1. The n must have zero charge as the charge is already conserved without neutrino.
2. Because the maximum energy carried away by electrons is equal to the maximum available energy or end point energy, the neutrino must have zero or almost zero rest mass.
3. The conservation of angular momentum requires the neutrino to have intrinsic spin as $\frac{1}{2}\hbar$.
4. A neutrino does not cause any appreciable amount of ionization in the medium when it is moving, so it passes undetected. This means that the neutrino has a very weak interaction with matter. Actually, it has none of the electromagnetic properties.

Further, it has also been proposed that there are two types of neutrinos—one which is ejected in the decay of neutron and second ejected in the decay of proton as under:

$$n \rightarrow p + e^- + \bar{\nu}$$
$$p \rightarrow n + e^+ + \nu$$

These two types of neutrinos are same or different. This question can be answered if we study the following reactions:

$$\bar{\nu} + p \rightarrow n + e^+$$
$$\nu + n \rightarrow p + e^-$$

These reactions are sometimes called inverse *b*-decay. It has been estimated that the probability of these reactions is very small, i.e. ~ 10^{-46} barn (1 barn = 10^{-28} m^2), compared to the probability of normal reactions which is ~ 1 barn.

3.6.6 Detection of Antineutrino ($\bar{n}$)

To show the existence of $\bar{n}$, experiments were started by Reines and Cowan in 1952 and completed in 1960. They used nuclear reactor as a source of $\bar{n}$. In nuclear reactors fission of
^{235}U produces neutron-rich fission products, which undergo b^--decay and emit $\bar{n}$. The flux of $\bar{n}$ was estimated to be ~ 10^{13} per cm^2 per second around the reactor. If the are real particles, then they should interact with protons as under:

$$\bar{n} + p \square n + e^+$$

For proton target Reines and Cowan took large quantities of water (~ 0.3 m^3) in which small quantity of $CdCl_2$ was dissolved. When $\bar{n}$ interact with proton (from water) a *n* and b^+ are produced. The positron b^+, will quickly encounter an electron and will annihilates giving two
g- rays of 511 keV each moving in opposite directions to each other to conserve momentum. This annihilation process takes 10^{-14} to 10^{-15} seconds. The neutrons produced collide with mostly hydrogen present in water and keep on losing energy and gradually slow down. This slowing down process takes about 8–10 □s. These slow neutrons are captured by Cd, forming a compound nucleus, which quickly decays emitting a *g*-ray of 9.1 MeV. Both these *g*-rays of 511 keV and 9.1 MeV are detected by scintillation detector giving light flashes. Electronic circuit picks up these

light flashes and gives electrical pulses. Thus, the characteristic signal for the presence of a $\bar{n}$ will be an electrical signal due to 511 keV *g*-ray followed about 10 □s later by electrical signal due to 9.1 MeV *g*-ray. With the experimental set-up developed by Reines and Cowan, they were able to observe about two such events per hour. After many years of ingenious and painstaking series of experiments, Reines and Cowan in 1960 were able to confirm directly the existence of $\bar{n}$.

3.6.7 Detection of Neutrino(*n*)

Another series of experiments were performed by Davis in 1955 to show that *n* and $\bar{n}$ are different particles. He demonstrated that $\bar{n}$ capture by neutron is not possible using the reaction

$$\bar{\nu} + {}^{37}_{17}\mathrm{Cl} \rightarrow {}^{37}_{18}\mathrm{Ar} + e^- \text{ or } \bar{\nu} + n \rightarrow p + e^-$$

If *n* are different from $\bar{n}$, then this reaction is not possible.
To prove that this reaction is not possible, Davis placed ~ 1000 gallons of CCl_4 in front of large flux of $\bar{n}$. The source of $\bar{n}$ in this experiment is again a nuclear reactor. If this capture of $\bar{n}$ by *n* is possible, then $^{37}_{18}$ Ar will be formed which is radioactive. By periodically searching for $^{37}_{18}$ Ar produced, he found no traces of $^{37}_{18}$ Ar. He concluded that reaction was not observed indicating that *n* and $\bar{n}$ are different particles.
Experiments for direct detection of *n* were carried out by Davis in 1955 and they lasted for about 10 years. These experiments were done in a mine which was 1500 metres deep. He used neutrinos emitted by sun as sun is a strong source of neutrinos because a large number of proton rich nuclei are formed during fusion process in sun. We know that proton-rich nuclei decay by b^+ and *n*. These experiments were based on the reaction

$$n + {}^{37}_{17}\mathrm{Cl} \square {}^{37}_{18}\mathrm{Ar} + e^- \text{ or } n + n \square p + e^-$$

He was able to detect 1 atom of $^{37}_{18}$ Ar per day. This $^{37}_{18}$ Ar formation confirms the presence
of *n*.

3.7 GAMMA DECAY

Alpha and beta decays of a radioactive nucleus usually leave the daughter nucleus in an excited state. If the excitation energy available with the daughter nucleus is not sufficient for further particle emission, it loses its excess energy by the following processes:

1. Gamma decay.
2. Internal conversion.
3. Internal pair conversion.

Gamma decay takes place more often than the other two processes.

3.7.1 Gamma Decay

Suppose a nucleus is in higher excited state with energy E_i, it makes a transition to lower energy state E_f or loses energy ΔE

$$h\nu = \Delta E = E_i - E_f \qquad (3.71)$$

If E_f is ground state of the nucleus, no further emission of γ-ray photons will take place, otherwise the nucleus will emit one or more γ-ray photons before reaching to ground state. During γ-decay mass and charge of the decaying nucleus does not change. For example, consider ^{137}Cs nucleus. It emits

β^--particles and the resulting nucleus ^{137}Ba is left in excited state, which emits a γ-ray of 662 keV energy. This total decay is represented by a diagram called the decay scheme of ^{137}Cs which is shown in Figure 3.23.

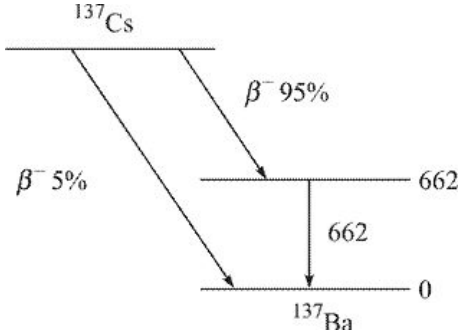


Figure 3.23 Decay scheme of ^{137}Cs.

About 95% of ^{137}Cs nuclei decay by b^{-}-emission to the first excited state with energy 662 keV of ^{137}Ba. Remaining 5% of ^{137}Cs nuclei decays by b^{-}-emission to the ground state of ^{137}Ba. The excited state of ^{137}Ba which is denoted by putting * on Ba as $^{137}Ba^{*}$ decays to ground state by emission of 662 keV *g*-ray. This *g*-decay is also written as

$$^{137}Ba^{*} \square\ ^{137}Ba + g$$

Similarly, the decay scheme of ^{60}Co is as shown in Figure 3.24.

$$^{60}Ni^{*} \square\square ^{60}Ni \square + g$$

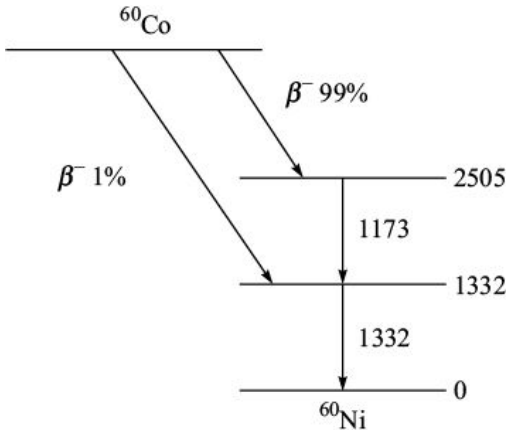


Figure 3.24 Decay scheme of ^{60}Co.

Remember *g*-decay is a two-body problem, i.e. products are the decayed nucleus and a *g*- ray. Therefore, the *g*-ray spectrum is a line spectrum or discrete spectrum.

Properties of Gamma Rays

- Gamma rays are electromagnetic waves just like X-rays or visible light.
- They always travel with velocity of light irrespective the energy of *g*-rays.
- They do not cause any appreciable ionization and are not deflected by electric or magnetic fields.
- Gamma-rays have high penetrating power, much larger than that of *a*- or *b*-particles.

It is interesting to mention here that both X- and *g*-rays are electromagnetic radiations and are identical. The differences between the two are shown in Table 3.2.

TABLE 3.2 Differences between X-rays and *g*-rays

S.No.	X-rays	*g*-rays
1.	They originate from outside the nucleus.	They originate from inside the nucleus.
2.	X-rays are emitted when electrons make a transition from higher shell to lower shell.	*g*-rays are emitted when nucleons inside the nucleus makes a transition from higher shell to lower shell.
3.	Generally, energy of X-rays is low, of the order of keV.	Energy of *g*-ray is high, of the order of MeV.

3.7.2 Internal Conversion

The excited nucleus can perform transition to a lower state either by emitting a *g*-ray or by transferring its energy directly to electrons surrounding the nucleus.

This process in which nucleus directly transfers its energy to one of the orbital electrons, by which the orbital electron is ejected from the atom, is known as *Internal Conversion*. These ejected electrons are known as conversion electrons.

The kinetic energy E_e of the conversion electron is given by

$$E_e = Eg - I_{be}$$

Here Eg is the energy available with the nucleus and I_{be} is the binding energy of the electron in a particular shell. Thus, the kinetic energy of the conversion electrons is unique. *Since the energy of the conversion electrons is unique, therefore, the energy spectrum of conversion electrons is line spectrum.* This line spectrum is superimposed on the continuous electron spectrum ejected in *b*-decay as shown in Figure 3.22.

Quantum mechanically this effect can be explained as under:

The nucleus and the orbital electrons being quantum mechanical systems have wave functions associated with them. The wave function of electron for a certain fraction of time overlaps the wave function of the nucleus. Thus, there is a finite probability that nucleus instead of *g*-emission transfers its energy to the orbital electron. The *K*-shell electron being nearest to the nucleus, so the probability of energy transfer to *K*-shell

electron will be maximum and next comes the *L*-shell electron, and so on. Because internal conversion and *g*-decay are competing processes and if lg and l_e are the transition probabilities for *g*-decay and internal conversion, then the total transition probability l is given by

$$l = lg + l_e$$

If the conversion electrons are ejected from *K*, *L* and *M* shells, then

$$l = lg + l_K + l_L + l_M$$

If a given radioactive material emits Ng rays in a given interval of time and N_e conversion electrons are also ejected in the same time, then we define

$$\alpha = \frac{N_e}{N_\gamma} = \frac{\lambda_e}{\lambda_\gamma}$$

Here a is called Internal Conversion Coefficient (ICC). If we record all the electrons (from *K*, *L*, *M*, etc. shells), the a is denoted by a_T (total internal conversion coefficient). However, if N_K, N_L, N_M, etc. are electrons ejected from *K*, *L*, *M*, etc. shells, then

$$\alpha_T = \frac{N_K + N_L + N_N + \cdots}{N_\gamma} = \frac{N_K}{N_\gamma} + \frac{N_L}{N_\gamma} + \frac{N_M}{N_\gamma} + \cdots$$

Therefore,

$$\alpha_T = \alpha_K + \alpha_L + \alpha_M + \cdots$$

Here a_K, a_L, a_M, etc. are called *K*, *L*, *M*, etc. conversion coefficients. The value of a lies between 0 and □.

3.7.3 Internal Pair Conversion

In addition to *g*-emission and internal conversion, there is another competing process—Internal Pair Conversion for the de-excitation of the nucleus. We have already seen that if the excitation energy of *g*-ray is greater than 1.022 MeV, this *g*-ray is absorbed in the medium and an electron– positron pair is created. It has been shown by Oppenheimer and Nedelsky in 1933 that if the excitation energy of the nucleus is greater than 1.022 MeV ($> 2m_0c^2$), the nucleus may de-excite by the creation of electron–positron pair somewhere in its own Coulomb field. Though

internal pair conversion can take place anywhere in the Coulomb field of the nucleus, but its probability is greatest at a distance from the nucleus which is of the order of $^{27}_{13}\mathrm{Al} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{30}_{15}\mathrm{P} + {}^{1}_{0}\mathrm{n}$ times the radius of the *K* shell of the atomic electrons.

Such a process in which nucleus de-excites by ejecting an electron–positron pair in its own Coulomb field is known as *Internal Pair Conversion*. This process is a competing process with *g*-emission and internal conversion.

Just like *g*-emission and internal conversion, internal pair conversion is due to electromagnetic interaction. Total excitation energy E_0 available with the nucleus is consumed as

$$E_0 = 2m_0c^2 + E_+ + E_-$$

where E_+ and E_- are the kinetic energies carried by positron and electron respectively and $2m_0c^2$ is the sum of the rest masses of the positron and electron. This process was explained theoretically by Rose in 1950. He found that

1. the probability of internal pair conversion increases with increasing the energy of excitation of the nucleus.
2. the probability of internal pair conversion decreases with the increase in multipolarity *L* (orbital angular momentum) of the transition.
3. the internal pair creation very slowly decreases with increasing *Z*.
4. if the excitation energy of the nucleus is very high (10 MeV), internal pair conversion becomes independent of multipolarity *L* of the transition.

These theoretical facts of Rose were nicely verified by experiments.

Differences between Pair Production and Internal Pair Conversion

- In pair production the excited nucleus emits a *g*-ray (of energy 1.022 MeV). This *g*-ray while moving in matter undergoes pair production in the Coulomb field of the electron or nucleus.

- In internal pair conversion, the excited nucleus does not emits a *g*-ray, but directly emits an electron–positron pair in its own Coulomb field.

Differences between Internal Conversion and Internal Pair Conversion

- In internal conversion, the excited nucleus transfers its excess energy to one of the orbital electrons and this electron is ejected from the atom. As there is vacancy in inner shell, an electron from higher shell makes a transition to the inner shell and thereby emitting a characteristic X-ray.
- In internal pair conversion, the excited nucleus emits a electron positron pair only. No
  X-ray is emitted as there is no vacancy in any atomic shells.

Differences between Internal Conversion and Electron Capture

- In internal conversion, the excited nucleus transfers its excess energy to one of the orbital electrons and this electron is ejected from the atom. As there is vacancy in
  inner shell, an electron from higher shell makes a transition to inner shell and thereby emitting a characteristic X-ray. The nucleus undergoing internal conversion remains the same.
- In electron capture, the nucleus captures an orbital electron. Inside the nucleus, the captured electron combines with a proton to form a neutron. Therefore, *Z* of the nucleus decreases by 1.
- Changes in mass and atomic numbers for various decay modes are summarized in
  Table 3.3.

TABLE 3.3 Variation in atomic mass and atomic number in different decays

Parameter	*a*-decay	*b*-decay			*g*-decay		
		b^--decay	b^+-decay	Electron capture	*g*-decay	Internal conversion	Internal pair conversion
Atomic no.	Decreases by 2 units	Increases by 1 unit	Decreases by 1 unit	Decreases by 1 unit	Remains same	Remains same	Remains same
Mass no.	Decreases by 4 amu	Remains same	Remains same	Remains same	Remains same	Remains same	Remains same

3.8 ARTIFICIAL OR INDUCED RADIOACTIVITY

Stable isotopes have fixed number of protons and neutrons in the nucleus. For example, sodium has 11 protons and 12 neutrons in its nucleus. When we bombard such stable isotopes with neutrons or high-energy charged particles like *p*, *d*, *a* or heavy ions, these particles enter the nucleus and disturb the stable balance of protons and neutrons. In some cases, this new balance constitutes a stable nucleus. But in majority of cases, the new system formed is unstable and it begins to emit *a*-particles, *b*-particles or *g*-rays like natural radioactive isotopes.

This process of inducing radioactivity in a given stable isotope by bombarding it with some suitable particles of proper kinetic energy is known as *artificial* or *induced activity*.

In the course of a study of the effect of *a*-particles from the naturally occurring radioelement Polonium, on the nuclei of certain light elements, B, Mg and Al in particular, I. Joliot Curie and her husband F. Joliot found that in addition to protons, there was an emission of neutrons and positrons. This result was not at all extraordinary since the nuclear process appeared to be essentially of the (*a*, *p*) type discovered by Rutherford, and a neutron and a positron could be together regarded as the equivalent of a proton. But earlier in 1934, the surprising fact was noted that although the formation of protons and neutrons ceased when the source of *a*-particles was removed, the positrons continued to be emitted. The foil of Al continued to emit positrons and the emission of positrons decays exponentially as in case of an ordinary or naturally occurring radio element. The same phenomenon was observed with B and Mg. The transmutation of B, Mg and Al by alphas has given birth to new radio elements emitting positrons.

According to Joliot, the reaction was (*a*, *n*) reaction as

$$^{27}_{13}\mathrm{Al} + ^{4}_{2}\mathrm{He} \rightarrow ^{30}_{15}\mathrm{P} + ^{1}_{0}\mathrm{n}$$

The $^{30}_{15}\mathrm{P}$ isotope of P obtained in this manner, which is not found in nature, would then be a radioactive species which decays by emission of positrons as under

$$^{30}_{15}\mathrm{P} \rightarrow ^{30}_{14}\mathrm{Si} + ^{0}_{+}e$$

The final product $^{30}_{14}Si$ is stable naturally occurring isotope of Si. With this discovery, artificial radioactivity came into existence. In this experiment artificial radioactive nucleus formed is $^{30}_{15}P$.

$$^{4}_{2}He + ^{27}_{13}Al \rightarrow ^{30}_{15}P + n$$
$$^{30}_{15}P \rightarrow ^{30}_{14}Si + ^{0}_{+1}e + \nu$$

These artificial radioisotopes once produced decay like natural radioisotopes.

Though the first radioisotope produced was by bombarding *a*-particles from polonium, but nowadays, we have a large number of particle accelerators and nuclear reactors at our disposal, which are used for producing radioisotopes. Radioisotopes in a reactor are produced mostly by (*n*, *g*) reaction. Some examples are:

$$^{23}_{11}Na + ^{1}_{0}n \rightarrow ^{24}_{11}Na + \gamma$$
$$^{59}_{27}Co + ^{1}_{0}n \rightarrow ^{60}_{27}Co + \gamma$$
$$^{109}_{47}Ag + ^{1}_{0}n \rightarrow ^{110}_{47}Ag + \gamma$$

Radioisotopes can also be produced by a variety of nuclear reactions in particle accelerators. Some such examples are:

$$^{58}_{28}Ni + ^{1}_{1}H \rightarrow ^{58}_{29}Cu + ^{1}_{0}n$$
$$^{114}_{48}Cd + ^{2}_{1}H \rightarrow ^{115}_{48}Cd + ^{1}_{1}H$$
$$^{27}_{13}Al + ^{14}_{7}N \rightarrow ^{38}_{19}K + ^{1}_{1}H + 2^{1}_{0}n$$

Today we have about 4000 artificial radioisotopes. These radioisotopes have found several uses in different branches of pure sciences, agriculture, medical sciences, etc. Some of these applications are discussed in the next section.

3.9 APPLICATIONS OF RADIOACTIVITY

Applications of radioactivity are varied and interesting. Before discussing these applications, we discuss some of the well-known cases where radioactivity has been utilized.

- Napoleon I Bonaparte, Emperor of the France died in 1821 at St. Helena. The exact cause of his death was not known at that time. It was only recently that by analyzing various elements present in his one scalp hair using radioactive technique, scientists were able to confirm that he died of repeatedly arsenic poisoning.
- In 1963 American President J.F. Kennedy was shot dead. It was not clear whether one or two assassins shot him because there were two bullet marks in his body. It was only by analyzing the bullet pieces recovered from his body using radioactive technique, it was confirmed that he was shot dead by a single assassin.
- Dinosaurs got extinct from the earth about 65.5 million years ago. The asteroid collision theory to account for the extinction of dinosaurs, which was first proposed by Walter Alvarez in the late 1970s, links the extinction event at the end of the Cretaceous period to a asteroid impact approximately 65.5 million years ago. Alvarez proposed that a sudden increase in iridium levels, recorded around the world in the period's rock stratum, was direct evidence of the impact. The bulk of the evidence now suggests that a 10 km-wide asteroid hit in the vicinity of the Yucatán Peninsula, creating the 170 km-wide Chicxulub Crater and triggering the mass extinction. The Yucatán Peninsula separates the Caribbean Sea from the Gulf of Mexico. The impact caused extinctions both directly (by heat from the meteorite impact) and also indirectly (via a worldwide cooling brought about when matter ejected from the impact crater reflected thermal radiation from the sun).
- We are familiar with un-manned space missions to moon and mars. In these missions one of the purpose is to know the elemental composition in the planets soil. For this purpose these flights are equipped with *a*-emitting radioactive source. The soil of the planet is exposed to *a*-particles. By measuring the energy of the back-scattered *a*-particles, scientists are able to get a clear picture of the elements present in a planet soil.

Having discussed some of the interesting applications of radioactivity, we discuss below some of the typical applications of radioactivity in various

fields.

3.9.1 Medicine

Radioisotopes find wide applications in medicines. They are used both for diagnostic and therapeutic purposes.

For diagnostic purposes, X-rays are most common example, which locate any fractured bone inside the body. X-rays are also used to study the gastrointestinal tract. ^{131}I and most recently ^{132}I and ^{123}I are used to study malfunctioning thyroid gland. Kidney function is also studied using compound containing ^{131}I. Tumours in the brain are located by injecting intravenously ^{99m}Tc and then scanning the head with suitable scanners.

A more recent development is Positron Emission Tomography (PET), which is a more precise and accurate technique for locating tumours in the body. A positron emitting radionuclide (e.g. ^{18}F) is injected to the patient and it accumulates in the target tissue. As it emits positrons which promptly combine with nearby electrons, it results in the simultaneous emission of two
g-rays in opposite directions. These *g*-rays are detected by a PET camera and give precise indication of their origin, i.e. depth also. This technique is also used in cardiac and brain imaging.

The most common therapeutic use of radioisotopes is in treating cancer. In this technique, cancerous cells are destroyed using *g*-rays from a ^{60}Co radioactive source. Sometimes wires or sealed needles containing radioactive isotope such as ^{192}Ir or ^{125}I are directly placed into the cancerous tissue. The radiations from the radioisotope attack the tumour as long as the needle/ wire is in place. When the treatment is complete, these are removed. This technique is frequently used to treat mouth, breast, lung and uterine cancer.

^{131}I is used to treat thyroid for cancers and other abnormal conditions of thyroid. ^{32}P is used to treat excess of red blood cells produced in the bone marrow.

3.9.2 Food and Agriculture

One of the more controversial uses of radiation today is food irradiation. High doses of
g-radiation do not make food radioactive. Gamma Irradiation kills bacteria, insects, and parasites, and retards spoilage in some foods like potatoes, onions, cooked food, etc. Irradiated foods are regularly eaten by astronauts on space missions, as well as by hospitalized patients with weak immune systems who need extra protection from microorganisms in food.

The irradiation process involves exposing food to intense controlled amounts of gamma rays from ^{60}Co or ^{137}Cs, X-rays, or electron beams from particle accelerators. The process has about the same effect on food as canning, cooking, or freezing. It kills pests and extends shelf life, but also reduces the food's nutritional value somewhat by destroying vitamins A, B1 (thiamin), C, and E. No radiation remains in the food after treatment.

3.9.3 Insect Control

About 10% of the world's crops are destroyed by insects. In efforts to control insect plagues, authorities often release sterile laboratory-raised insects into the wild. These insects are made sterile using ionizing radiation. They are irradiated with this radiation before they hatch. Female insects that mate with sterile male insects do not reproduce, and the population of the insect pests can be quickly curbed as a consequence. This technique of releasing sterile insects into the wild, called the *sterile insect technique* (SIT), is commonly used in protecting agricultural industries in many countries around the world.

3.9.4 Reactors

Atomic nuclei are storehouse of huge amounts of energy. This huge amount of energy can be explored when a heavy nucleus is broken into two or more lighter fragments or when two light nuclei are fused into a heavier one. Let us calculate energy released when we break a heavy nucleus into two lighter fragments.

If we bombard ^{235}U with neutron, the relevant nuclear reaction is:

$$^{235}U + {}^{1}n \rightarrow \text{fission products} + \text{neutrons} + \text{energy (}\sim 200\text{ MeV)}$$

The average number of neutrons emitted in this reaction is 2.43. Some of the neutrons in reaction can be used to induce fission in another ^{235}U nucleus, thus continuing a controlled, self-perpetuating nuclear chain reaction.

It has been estimated that 1 g of U or Pu on fission would produce about 1 MW of power for one day. The same energy production would require more than 3 tons of coal or about 2300 litres of fuel oil per day (see Section 4.7.2).

Other possible fission reactions are:

$$^{233}U + {}^{1}n \rightarrow \text{fission products} + \text{neutrons} + \text{energy (}\sim\text{200 MeV)}$$

$$^{239}Pu + {}^{1}n \rightarrow \text{fission products} + \text{neutrons} + \text{energy (}\sim\text{200 MeV)}$$

Similarly, if we fuse two lighter nuclei into a heavier one, then

$$^{2}H + {}^{3}H \rightarrow {}^{4}He + n + 17.6 \text{ MeV}$$

It has been estimated that if 0.5 g of ^{2}H and 0.5 g of ^{3}H is fused together it would produce about 5 MW of power for one day. The same energy production would require more than 15 tons of coal or about 12,000 litres of fuel oil for one day (see Section 4.8.1). This shows that huge amount of energy is stored in atomic nuclei. This energy is released in controlled manner in nuclear reactors to generate electricity and in uncontrolled manner in atomic and thermonuclear (hydrogen) bombs.

In India we have 19 commercial power reactors in operation and seven reactors under construction. They are shown in Tables 3.4 and 3.5.

TABLE 3.4 Commercial power producing reactors in India

S.No.	Reactor	Location	Type*	Capacity (MWe)+	Year of start
1.	TAPS-1	Tarapur, Maharashtra	BWR	160	1969
2.	TAPS-2	Tarapur, Maharashtra	BWR	160	1969
3.	TAPS-3	Tarapur, Maharashtra	PHWR	540	2006
4.	TAPS-4	Tarapur, Maharashtra	PHWR	540	2005
5.	RAPS-1	Rawatbhata, Rajasthan	PHWR	100	1973
6.	RAPS-2	Rawatbhata, Rajasthan	PHWR	200	1981
7.	RAPS-3	Rawatbhata, Rajasthan	PHWR	220	2000
8.	RAPS-4	Rawatbhata, Rajasthan	PHWR	220	2000
9.	RAPS-5	Rawatbhata, Rajasthan	PHWR	220	2010

10.	RAPS-6	Rawatbhata, Rajasthan	PHWR	220	2010
10.	RAPS-6	Rawatbhata, Rajasthan	PHWR	220	2010
11.	MAPS-1	Kalpakkam, Tamil Nadu	PHWR	220	1984
12.	MAPS-2	Kalpakkam, Tamil Nadu	PHWR	220	1986
13.	NAPS-1	Narora, Uttar Pradesh	PHWR	220	1991
14.	NAPS-2	Narora, Uttar Pradesh	PHWR	220	1992
15.	KAPS-1	Kakarpar, Gujarat	PHWR	220	1993
16.	KAPS-2	Kakarpar, Gujarat	PHWR	220	1995
17.	KAIGA-1	Kaiga, Karnataka	PHWR	220	2000
18.	KAIGA-2	Kaiga, Karnataka	PHWR	220	2000
19.	KAIGA-3	Kaiga, Karnataka	PHWR	220	2007

Total nuclear power plant capacity: 4560 MWe

*PHWR: Pressurized Heavy Water Reactor, BWR: Boiling Water Reactor.

$^{+}$MWe: megawatt electric.

TABLE 3.5 Commercial power producing reactors in India under construction

S.No.	*Reactor*	*Location*	*Type*	*Capacity* (MWe)	*Estimated year of start*
1.	KAPP-1	Kudankulam, Tamil Nadu	PWR	1000	2011
2.	KAPP-2	Kudankulam, Tamil Nadu	PWR	1000	2011
3.	RAPS-7	Rawatbhata, Rajasthan	PHWR	700	2016
4.	RAPS-8	Rawatbhata, Rajasthan	PHWR		

700	2016				
5.	KAIGA-4	Kaiga, Karnataka	PHWR	220	2010
6.	KAPS-3	Kakarpar, Gujarat	PHWR	700	2015
7.	KAPS-4	Kakarpar, Gujarat	PHWR	700	2015

Total estimated nuclear power plant capacity: 5020 MWe

3.9.5 Research Reactors

Research reactors are essentially nothing more than a source of neutrons. They are much smaller in comparison to nuclear power reactors and, therefore, use less fuel. They pose less of a risk to the environment.

Neutrons produced by research reactors are used for experiments as well as the manufacture of artificial radioactive isotopes. These artificial radioisotopes can be used in various fields including medicine, industry and agriculture.

Hundreds of research reactors exist in the world. They are invaluable tools for research and development in science. In India we have nine research reactors and the various parameters of these research reactors are as shown in Table 3.6.

TABLE 3.6 Research reactors in India

Reactor	Year of start	Power (MWe)	Fuel	Moderator	Coolant	Neutron flux	Location
Apsara	1956	1	Enriched-U	H_2O	H_2O	10^{13}	BARC, Mumbai
Cirus	1960	40	Natural-U	D_2O	H_2O	6.7 □ 10^{13}	BARC, Mumbai
Zerlina*	1961	100	Natural-U	D_2O	D_2O	0.5 □ 10^{8}	BARC, Mumbai
Purnima-I	1972	–	PuO_2–Al	–	–	–	BARC, Mumbai
Purnima-II	1984	–	^{233}U(uranyl nitrate solution)	H_2O	–	–	BARC, Mumbai
Purnima-III	1990	–	^{233}U rods	H_2O	–	–	BARC, Mumbai
Dhruva	1985	100	Natural-U	D_2O	H_2O	1.8 □ 10^{14}	BARC, Mumbai
FBTR	1985	40	U–Pu carbide	–	Liquid sodium	–	IGCAR, Kalpakkam
Kamini	1996	30	^{233}U	H_2O	H_2O	10^{12}	IGCAR, Kalpakkam

*Decommissioned in 1983.

3.9.6 Geology and Element Identification

Radioactivity is used to identify the location of deposits of uranium and other radioactive minerals. This is useful in mining exploration. The intensity of detected radiation also is an indication of the amount of radioactive element that may be located there.

3.9.7 Radioactive Dating

The principle of radioactive decay is applied in the technique of *radioactive dating*, a process widely used by scientists to determine the age of materials and artifacts.

Radioactive dating is defined as the method of determining the age of biological or geological samples by using the radioactive technique.

There are many radioisotopes by which we can determine the age of a given object, but the two most commonly used methods are:

- Radiocarbon dating.
- Uranium dating.

Radiocarbon Dating

Radioactive ^{14}C atoms exist naturally in very minute quantities. ^{14}C is everywhere around us: in our clothes, in the food we eat, even in the air we breathe. The ratio of radioactive ^{14}C atoms to stable ^{12}C atoms in the atmosphere has remained constant over thousands of years. Although ^{14}C naturally decays, it is also continually being formed. ^{14}C atoms are formed when neutrons from the cosmic radiation collide with ^{14}N atoms in the atmosphere. Thus, the decay of ^{14}C is reasonably balanced with its production, resulting in a constant ratio of ^{14}C to ^{12}C.

Carbon dioxide (CO_2) molecules in the air contain both isotopes of carbon. This CO_2

is continually used by plants to grow. Because the ratio of ^{14}C to ^{12}C in atmospheric CO_2 is constant, the intake of CO_2 by a plant results in a

constant ratio of the two isotopes in the plant's body while it is alive. However, when the plant dies it no longer takes in CO_2. As a result, the ^{14}C decaying in the dead plant can not be replenished by a "fresh supply" of more CO_2, resulting in the decrease in activity of ^{14}C with time. Because animals eat plants and inhale air, the activity of ^{14}C decreases once they die, since the ^{14}C cannot be replenished. In order to determine the radioactivity of ^{14}C, a small portion of the sample is burnt, so that carbon present in it reacts with oxygen to form CO_2. The CO_2 that contains ^{14}C is radioactive, and the amount is measured using a radiation counter. Burning is done to facilitate measuring the level of ^{14}C. ^{14}C has a half-life of about 5730 years. This means that in a given sample of a carbon-containing substance, (without the ^{14}C being replenished) the activity of ^{14}C decreases by half every 5730 years. Suppose someone discovers an ancient manuscript and finds that the activity of ^{14}C in the paper is half of that found in living trees. This would mean that the manuscripts would be about 5730 years old.

The use of radioactive ^{14}C for dating was first done by William Libby, at the University of Chicago, USA, in 1947.

The age of a given sample can be determined using Eq. (3.7)

$$A = A_0 e^{-lt}$$

In calculating the age of a sample, we make the following assumption:

The activity of ^{14}C in a living sample has remained same over thousands of years.

In the above equations A_0 is the activity of the sample, when it is living. A is the activity of the sample whose age is to be determined. For example, if we wish to determine the age of a given piece of wood, A_0 is the activity for a fresh wood piece taken from a presently living tree. A is the activity of the sample whose age is to be determined.

The relatively short half-life of ^{14}C (5730 years) means that the amount of ^{14}C remaining in materials and objects older than about 30,000 years is too small to be measured experimentally. Thus, carbon dating has following two limitations:

- It cannot measure the age of an object older than about 30,000 years.
- It cannot measure the age of non-living objects like rocks and minerals.

Uranium Dating

Uranium dating method is applicable to ores and rocks, etc. We have seen in Section 3.2 that ^{238}U decays into ^{206}Pb with a half-life of $4.47 \square 10^9$ years.

Let N_0 be the amount of ^{238}U in a given sample, when it was initially formed. N_U is the amount of ^{238}U still left in the sample after the lapse of time t, which represents the age of the sample. All the ^{238}U, which has decayed, has been converted into ^{206}Pb. Let the quantity of ^{206}Pb, be N_{Pb}. The quantity of ^{238}U, when the sample was formed, is

$$N_0 = N_U + N_{Pb}$$

Therefore, we have Eq. (3.6) as

$$N = N_0 e^{-lt}$$

Here, $N_0 = N_U + N_{Pb}$ and $N = N_U$. Therefore,

$$N_U = (N_U + N_{Pb})e^{-lt}$$

Thus, knowing the amounts N_U (of ^{238}U) and N_{Pb} (of ^{206}Pb) and l (decay constant of ^{238}U) in a given sample, the age t of the sample can be calculated.

The limitations of this method are:

- It is applicable to rocks, minerals, etc. containing $^{238}U\square$only.

- This method is applicable to samples having age □ 10^9 years only.

Geologists measure the amounts of other radioisotopes such as ^{84}Rb and ^{40}K, etc. in rocks to determine their age. Measurements show that the oldest rocks on earth are about 4.6 billion years old, which is a reasonably accurate estimate of the earth's age. Similarly, analysis of fossilized plants shows that they first occurred on earth about 3 billion years ago.

3.9.8 Radioactive Tracers

Radioisotopes can be used to help understand chemical and biological processes in plants and other living beings. This is true for the following two characteristic properties.

- Radioisotopes are chemically identical with other isotopes of the same element and are substituted in chemical reactions.
- Radioactive forms of the element can be easily detected by *b* and *a* radiations emitted by them.

Therefore, these properties of the radioisotope, namely its radioactivity, can act as a tag or label, which permits the fate of the element or its compound containing this element to be traced through a series of chemical or physical changes. The element which has been labelled or tagged is called tracer element. For example, if in $CaPO_4$, ordinary phosphorus is replaced by ^{32}P, which is radioactive, then phosphorus in $CaPO_4$ is known as tracer element and this technique is known as tracer technique. The tracer technique finds wide applications in various fields. Some of the applications of tracer technique are discussed below.

Agriculture

To see how plants utilize a given fertilizer, a solution of phosphate containing radioactive ^{32}P is injected into the root system of a plant. Since ^{32}P behaves identically to that of ^{31}P, a more common and non-radioactive form of the element, it is used by the plant in the same way. A GM counter is then used to detect the movement of the radioactive ^{32}P throughout the

plant. This information helps scientists to understand the detailed mechanism of how plants utilize phosphorus to grow and reproduce.

Medicine

A **radioactive tracer**, or **radioactive label**, is a chemical compound in which one or more atoms have been replaced by a radioisotope. It finds wide applications in medicines. Radioisotopes of hydrogen, carbon, phosphorus, sulphur and iodine have been used extensively to trace the path of biochemical reactions. A radioactive tracer can also be used to track the distribution of a substance within a natural system such as in a cell or in a tissue. Radioactive tracers form the basis of a variety of imaging systems, such as positron emission tomography (PET) scans, single-photon emission computed tomography (SPECT) scans and technetium scans.

Since hydrogen atoms are present in all organic compounds, so tritium is frequently used as a tracer in biochemical studies. ^{14}C has been used extensively to trace the progress of organic molecules through metabolic pathways.

^{13}N, ^{15}O, ^{18}F etc. are used in PET scan.^{33}P is used in DNA sequencing.

In metabolism research, tritium and ^{14}C-labelled glucose are commonly used in glucose clamps to measure rates of glucose uptake, fatty acid synthesis and other metabolic processes.

Industry

Radioisotopes are commonly used in industry for checking blocked water pipes, detecting leakage in oil pipes, etc. For example, small quantity of radioactive ^{24}Na is placed in a small enclosed ball and is allowed to move in pipe with water. The moving ball containing radioisotope is monitored with a detector. If the movement of ball stops, indicating the blocked pipe. Similarly, radioisotope ^{24}Na is mixed with oil flowing in an underground pipe. With radiation detector, the radioactivity over the pipe is monitored. If there is a leakage at a particular place, the radiation detector will show large activity at that particular place.

3.9.9 Neutron Activation Analysis (NAA)

It is one of the most sensitive and nondestructive techniques used for qualitative and quantitative elemental analysis of multiple major, minor and trace elements in samples from almost every conceivable field of scientific or technical interest. For many elements, NAA offers sensitivities that are superior to those possible by any other technique. In this technique, a small sample of the material to be analysed is exposed to a high flux (~10^{13} n/cm^2/s) of thermal neutrons from a reactor. Nuclei of elements present in the given sample undergo (*n*, *g*) reaction and may become radioactive. For example, sodium present in the sample undergoes the following reaction:

$$^{23}_{11}\mathrm{Na} + {}^{1}_{0}n \rightarrow {}^{24}_{11}\mathrm{Na} + \gamma$$

$^{24}_{11}\mathrm{Na}$ so formed is radioactive. Many of such radioactive nuclei decay through *b* and subsequent by *g* emission. These *g*-rays are characteristic of the nuclei emitting *g*-rays. By precise measurement of the *g*-ray energies emitted by neutron exposed sample, helps us in determining the presence of elements in the sample. From the *g*-ray intensities, it can be determined, exactly how much of the original neutron capturing isotope is present and consequently the amount of that element present in the sample can be determined. The applications for NAA are practically limitless for solid, liquid or gaseous samples. Most of the elements can be measured with an accuracy of 1 part per million, ~20 elements can be measured with two parts per billion levels and about 5–6 elements with 1–2 parts per trillion levels.

3.9.10 In Space Flights

Radioisotopes find wide applications in space flights. Suppose a spacecraft is somewhere near Neptune, this craft lacks the warm touch of sunrays. Thus, solar panels cannot provide power. Chemical batteries providing necessary power for the spacecraft would be too heavy to carry on board. The question is how to get sufficient power for the spacecraft. A radioisotope thermoelectric generator (RTG) is an answer to this question. An RTG generates electric power by converting heat released during the nuclear decay of radioactive isotope into electricity. This device has been in use since 1956. For example, an RTG fabricated in 1959 weighing about 2

kg and providing power of 2.5 watts was capable of delivering 11.60 kW-h of electric energy over a period of 280 days. This is the energy produced by nickel–Cadmium batteries weighing about 320 kg. Almost all Appolo missions to moon, Viking missions to Mars, the Pioneer, Voyager, etc. missions to outer solar system carried RTG as a source of power. The radioisotopes commonly used in RTG are either ^{210}Po or ^{238}Pu. Recent mission to Mars in which Curiosity landed on Mars is powered by ^{238}Pu-based RTG providing 110 W of electricity.

3.9.11 In Sea

Nuclear power is particularly suitable for vessels which need to be at sea for long periods without refueling. These vessels (ships/submarines) have a small nuclear reactor on board, which produces steam to run turbines. These turbines directly drive the propellers as well as electrical generators. Today, a large number of nuclear-powered ships and submarines are cruising the oceans.

NUMERICAL PROBLEMS

Section 3.2

Solved Problems

1. How many curies are there in 10^{10} Bq?

Solution: We know that $3.7 \square 10^{10}$ Bq = 1 Ci
Therefore,

$$10^{10}\ \text{Bq} = \frac{1}{3.7 \times 10^{10}} \times 10^{10}\ \text{Ci}$$

$$= 0.27\ \text{Ci}$$

2. Calculate the activity of 10 g of ^{232} Th. Given: $l_{232Th} = 1.58 \square 10^{-18}\ s^{-1}$.

Solution: 1 g mole of ^{232}Th = 232 g of ^{232}Th

$$232 \text{ g of } {}^{232}\text{Th contains} = 6.023 \times 10^{23} \text{ atoms of } {}^{232}\text{Th}$$

$$10 \text{ g of } {}^{232}\text{Th contains} = \frac{6.023 \times 10^{23}}{232} \times 10 = 2.596 \times 10^{22} \text{ atoms}$$

Therefore,

$$N = 2.596 \times 10^{22}$$

$$\text{Activity } Nl = 2.596 \times 10^{22} \times 1.58 \times 10^{-18}$$

$$= 4.104 \times 10^{4} \text{ dps}$$

3. What will be the mass of 1 curie sample of a radioactive substance of atomic mass 214? Its half-life is 26.8 min.

Solution: Given:

$$\text{Activity } A = 1 \text{ Ci} = 3.7 \times 10^{10} \text{ dps}$$

$$\text{Half-life } t_{1/2} = 26.8 \text{ min} = 1608 \text{ s}$$

$$\text{Decay constant } \lambda = \frac{0.6931}{t_{1/2}}$$

$$= \frac{0.6931}{1608}$$

$$= 4.31 \times 10^{-4} \text{ s}^{-1}$$

Let m be the mass of sample having activity 1 Ci.

Therefore,

214 g of radioactive substance contains = 6.023×10^{23} atoms

m g of radioactive substance contains $N = \frac{6.023 \times 10^{23} \times m}{214}$ atoms

Now, we have

$$\text{Activity } A = l \times N$$

Therefore,

$$3.7 \times 10^{10} = 4.31 \times 10^{-4} \times \frac{6.023 \times 10^{23} \times m}{214}$$

or

$$m = \frac{3.7 \times 10^{10} \times 214}{4.31 \times 10^{-4} \times 6.023 \times 10^{23}}$$

$$= 3.05 \times 10^{-8} \text{ g}$$

4. The half-life of $^{238}_{92}$U against a-decay is 4.5×10^9 years. Find the activity of 1 kg of $^{238}_{92}$U.

Solution: Half-life $t_{1/2} = 4.5 \times 10^9$ years $= 1.419 \times 10^{17}$ s

Therefore,

$$\text{Decay constant } \lambda = \frac{0.6931}{t_{1/2}}$$

$$= \frac{0.6931}{1.419 \times 10^{17}}$$

$$= 4.88 \times 10^{-18} \text{ s}^{-1}$$

$$238 \text{ g of } ^{238}_{92}\text{U contains} = 6.023 \times 10^{23} \text{ atoms}$$

$$1000 \text{ g of } ^{238}_{92}\text{U contains } N = \frac{6.023 \times 10^{23} \times 1000}{238} \text{ atoms}$$

Now, we have

$$\text{Activity } A = l \times N$$

Therefore,

$$A = 4.88 \times 10^{-18} \times 2.53 \times 10^{24}$$

$$= 1.236 \times 10^7 \text{ dps}$$

$$= 3.34 \times 10^{-4} \text{ Ci}$$

5. One milligram of a radioactive material with half-life of 1600 years is kept for
2000 years. Calculate the mass, which would have decayed by this time.

Solution: Given:

$$\text{Half-life } t_{1/2} = 1600 \text{ years}$$

Therefore,

$$\text{Decay constant } \lambda = \frac{0.6931}{t_{1/2}}$$

$$= \frac{0.6931}{1600} = 4.332 \times 10^{-4} \text{ years}^{-1}$$

Now, we have from Eq. (3.6)

$$N = N_0 e^{-\lambda t} \text{ or } \frac{N_0}{N} = e^{\lambda t}$$

Therefore,

$$\frac{N_0}{N} = e^{4.332\times10^{-4}\times2000}$$
$$= e^{0.8663}$$
$$= 2.378$$

If N_0 is the initial number of nuclei at time t = 0, and the corresponding mass of the radioactive substance is m_0. Similarly, if N is the number of nuclei left at time t (= 2000 years), the corresponding mass be m, then we must have

$$\frac{N_0}{N} = \frac{m_0}{m}$$

or

$$\frac{N_0}{N} = \frac{m_0}{m} = 2.378$$

Given that m_0 = 1 mg
Therefore,

$$\frac{m_0}{m} = 2.378$$

or

$$m = \frac{1}{2.378} = 0.4205 \text{ mg}$$

Therefore, the amount of radioactive substance decayed is

$$= 1 - 0.4205 = 0.5795 \text{ mg}$$

6. The activity of a certain nuclide decreases to 15% of its original value in 10 days. Find its half-life.

Solution: Given:

$$t = 10 \text{ days}$$

Let the initial activity = A_0

Activity left after 10 days = $\frac{15}{100} A_0$

We have from Eq. (3.7) $A = A_0 e^{-l\square t}$

Therefore,

$$\frac{15}{100}A_0 = A_0 e^{-l\square 10}$$

or

$$e^{-10\square l} = 0.15$$

or

$$e^{10\square l} = \frac{1}{0.15} = 6.6667$$

or

$$10 \square\, l = \log_e 6.6667 = 1.8971$$

Therefore,

$$l = 0.18971 \text{ day}^{-1}$$

$$t_{1/2} = \frac{0.6931}{\lambda}$$
$$= \frac{0.6931}{0.18931}$$
$$= 3.65 \text{ days}$$

7. Calculate the mass in kg of 1 curie and 1 rutherford in a sample of ^{226}Ra, whose half- life is 1620 years.

Solution: Given:

Half-life $t_{1/2}$ = 1620 years

Therefore,

$$\text{Decay constant } \lambda = \frac{0.6931}{t_{1/2}}$$
$$= \frac{0.6931}{1620 \times 365 \times 24 \times 3600}$$
$$= 1.3567 \times 10^{-11} \text{ s}^{-1}$$
$$\text{Activity } A = 1 \text{ Ci} = 3.7 \times 10^{10} = N\lambda$$

or

$$3.7 \times 10^{10} = N \times 1.3567 \times 10^{-11}$$

or

$$N = \frac{3.7 \times 10^{10}}{1.3567 \times 10^{-11}}$$
$$= 2.727 \times 10^{21} \text{ atoms}$$

$6.023 \square 10^{26}$ atoms have mass = 226 kg

$2.272 \square 10^{21}$ atoms have mass = $\frac{226 \times 2.727 \times 10^{21}}{6.023 \times 10^{26}} = 1.023 \square 10^{-3}$ kg

Therefore, the mass in kg of 1 curie in a sample of $^{226}Ra = 1.023 \square\square 10^{-3}$ kg

$$\text{Activity } A = 1R = 10^6 = Nl$$

or

$$10^6 = N \square 1.3567 \square 10^{-11}$$

or

$$N = \frac{10^6}{1.3567 \times 10^{-11}} = 7.37 \times 10^{16} \text{ atoms}$$

6.023×10^{26} atoms have mass = 226 kg

$$7.37 \times 10^{16} \text{ atoms have mass} = \frac{226 \times 7.37 \times 10^{16}}{6.023 \times 10^{26}} = 2.77 \times 10^{-8} \text{ kg}$$

Therefore, the mass in kg of 1 rutherford in a sample of $^{226}Ra = 2.77 \square 10^{-8}$ kg.

8. If a radioactive material initially contains 3.0 mg of ^{234}U, how much it will contain
after 1,50,000 years? What will be its activity at the end of this time? Given:

$t_{1/2} = 2.5 \square 10^5$ years and $l = 8.8 \square 10^{-14}$ s^{-1}

Solution: 234 g of ^{234}U contains = $6.023 \square 10^{23}$ atoms

$3 \square 10^{-3}$ g of ^{234}U contains $N = \frac{6.023 \times 10^{23} \times 3 \times 10^{-3}}{234} = 7.721 \square 10^{18}$ atoms

Number of atoms left after 1,50,000 years is given by

$$N = N_0 e^{-\lambda \times t}$$

Now, $$\lambda = \frac{0.6931}{t_{1/2}}$$

$$= \frac{0.6931}{2,50,000}$$

$$= 2.7724 \square 10^{-6} \text{ years}^{-1}$$

Therefore,

$$N = 7.721 \square\square 10^{18} e^{-2.7724 \square 10-6 \square 150,000}$$

$$= 7.721 \square 10^{18} e^{-0.41586}$$

$$= 7.721 \square 10^{18} \square 0.65977$$

$$= 5.094 \square 10^{18} \text{ atoms}$$

$6.023 \square 10^{23}$ atoms have mass = 234 g

$5.094 \square 10^{18}$ atoms have mass = $\frac{234 \times 5.094 \times 10^{18}}{6.023 \times 10^{23}}$ = 1.979 mg

Therefore, the weight of the sample left after 1,50,000 years will be 1.979 mg.

$$\text{Activity } Nl = 5.094 \square 10^{18} \square 8.8 \square 10^{-14}$$

$$= 448,272 \text{ dps}$$

$$= 1.212 \square 10^{-5} \text{ Ci}$$

$$= 0.01212 \text{ mCi}$$

$$= 12.12 \square \text{Ci}$$

9. Calculate the activity of ^{40}K in a human body weighing 100 kg. Assume that 0.35% of the body weight is potassium. In natural potassium, the abundance of ^{40}K is 0.012%. Half-life of $^{40}K = 1.31 \square 10^9$ years.

Solution: Total weight of potassium in 100 kg human body = $100 \square 0.35 \square 10^{-2} = 0.35$ kg

Weight of ^{40}K with abundance 0.012% = 0.35 □ 0.012 □ 10^{-2} = 4.2 □ 10^{-5} kg

From Avogadro's hypothesis, number of atoms in 4.2 □ 10^{-5} kg of ^{40}K

$$N = \frac{6.023 \times 10^{26}}{40} \square 4.2 \square 10^{-5}$$

$$= 6.324 \square 10^{20} \text{ atoms}$$

Therefore, activity of $^{40}K = l \square N = \frac{0.693 \times N}{l}$

$$= \frac{0.693 \times 6.324 \times 10^{20}}{1.31 \times 10^{9} \times 365 \times 24 \times 60 \times 60}$$

$$= 1.061 \square 10^{4} \text{ dps}$$

Activity in human body = 1.061 □ 10^{4} dps

= 0.287 □Ci

10. Radionuclides are useful source of small amounts of energy in space vehicles, remote communication systems etc. One such radionuclide is ^{210}Po. Calculate the power available in watts from a 10 mg of ^{210}Po. It is an *a*-emitter with an energy of 5.3 MeV. Given $t_{1/2}$ = 138 days and m_{Po} = 209.982848 amu.

Solution: 210 g of ^{210}Po contains = 6.023 □ 10^{23} nuclei.

10 □ 10^{-3}g of ^{210}Po contains $\frac{6.023 \times 10^{26}}{210} \times 10 \times 10^{-3}$ nuclei.

$$N = 2.87 \square 10^{19} \text{ nuclei}$$

Decay constant $l = \frac{\ln 2}{t_{1/2}} = \frac{0.6931}{138 \times 24 \times 3600} = 5.81 \times 10^{-8} s^{-1}$

Therefore,

Activity = $N \square l$ = 2.87 □ 10^{19} □ 5.81 □ 10^{-8} = 1.667 □ 10^{12} dps.

1 disintegration/s produces power = 5.3 MeV/s.

1.667×10^{12} dps will produce power = $5.3 \times 1.667 \times 10^{12}$ MeV/s

$$= \frac{5.3 \times 1.667 \times 10^{12} \times 1.6 \times 10^{-6}}{10^{7}} = 1.4 \text{ W}$$

Unsolved Problems

1. How many Bq are there in 2700 disintegrations per second? [**Ans.** 2700 Bq]
2. How many Bq are there in 95 mCi? [**Ans.** 3.515×10^{8} Bq]
3. What is the activity of 3.5×10^{10} atoms of ^{60}Co? l for ^{60}Co = 0.132 year^{-1}. [**Ans.** 146 dps]
4. How many atoms are there in 1.2×10^{10} dps of ^{54}Co? Half-life of ^{54}Co = 1.46 min. [**Ans.** 1.51×10^{12} atoms]
5. A ^{131}I source has an activity of 30 mCi today. What was the activity of the source one month ago? Given: l = 0.086 d^{-1}. [**Ans.** 0.395 Ci]
6. A ^{88}Rb source has an activity of 1200 Bq. What is the activity of the source 30 minutes back? Given: l = 0.039 min^{-1}. [**Ans.** 3866 dps]
7. The decay constant for ^{133}Xe is 0.13 d^{-1}. What is its half-life and average life? [**Ans.** 5.38 days, 7.69 days]
8. A radioisotope decays in such a manner that after 1 hour, only 25% of the initial amount remains. Calculate the decay constant and half-life of the isotope. [**Ans.** 1.386 hour^{-1}, 30 min]
9. How much time will it take 100 mCi of ^{3}H to decay to 25 mCi. Given: l = 0.056 year^{-1}. [**Ans.** 24.75 years]
10. The half-life of a-emitter ^{210}Po is 138 days. What mass of ^{210}Po is needed for a 10 mCi source? [**Ans.** 2.22×10^{-9} kg]
11. A beam of thermal neutrons having energy 0.025 eV passes through an apparatus of

    50 cm long at the rate of 10^{9} s^{-1}. $t_{1/2}$ for neutrons is 12 min. Calculate

the rate at which neutrons decay are expected in the apparatus. [**Ans.** ~220 neutrons/s]

12. Find the number of *a*-particles emitted per second by 3 mg of ^{226}Ra. Also find the activity in microcuries. $t_{1/2}$ for ^{226}Ra = 1620 years. [**Ans**. 1.1 □ 10^{8}, 3000 □Ci]

13. Estimate the number of *a*-particles emitted per second by 2 cc of ^{222}Rn at NTP. Also estimate the activity in millicuries. $t_{1/2}$ for ^{222}Rn = 3.82 days. [**Ans.** 1.13 □ 10^{14}, 3 □ 10^{6} mCi]

14. Calculate the amount of ^{210}Po to give us an *a*-particle source of 5 mCi. [**Ans.** 1.1 □g]

15. Find the number of *b*-particles emitted per minute by natural KCl of 0.2 kg. Given that 90% of ^{40}K disintegrations results in *b*-emission and the abundance of ^{40}K in natural potassium is 0.0117%. [**Ans.** 9.12 □ 10^{4}]

16. If one starts with 1 kg of ^{3}H, how long will it be until only 1 g is left? Half-life of ^{3}H is 12.26 years. [**Ans.** 122.13 years]

17. The activity of ^{32}P today is 50 □Ci. What will be the activity in 6 weeks from now? Given: *l* for ^{32}P = 0.0495 d^{-1}. [**Ans.** 6.25 □Ci]

18. What is the fraction of atoms left after 10 half-lives of a substance? [**Ans.** 1/1024]

19. Suppose you begin with 1.0 □ 10^{-2} g of a pure radioactive substance and 4 h later determine that 0.25 □ 10^{-2} g remain. What is the half-life of the substance? [**Ans.** 2h]

20. The half-life of ^{32}P is 14.3 days. What interval of time is required for 100 mCi of ^{32}P to decay to 25 mCi? What time is required for decay of 7/8 of the ^{32}P atoms? [**Ans.** 42.9 days]

21. What is the mass in grams of 1000 MBq of pure ^{32}P? How many ^{132}I atoms constitute 1000 MBq? What is the mass in grams of 1000 MBq

of Na_3PO_4 if all the phosphorus in the compound is radioactive? [**Ans.** 9.47 □ 10^{-8}g, 1.78 □ 10^{15} atoms, 4.88 □ 10^{-7}g]

22. How many MBq of ($t_{1/2}$ = 2.3 h) should be ordered so that the sample activity will be 500 MBq when it arrives 24 h later? [**Ans.** 6,91,733 MBq]

Section 3.3

Solved Problems

1. Complete the following reactions by putting appropriate particle on the arrow:

$$^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} \rightarrow {}^{234}_{91}\text{Pa} \rightarrow {}^{234}_{92}\text{U}$$
$$^{211}_{82}\text{Pb} \rightarrow {}^{211}_{83}\text{Bi} \rightarrow {}^{207}_{81}\text{Tl} \rightarrow {}^{207}_{82}\text{Pb}$$
$$\downarrow\rightarrow {}^{211}_{84}\text{Pa} \rightarrow\uparrow$$

Solution:

$$^{238}_{92}\text{U} \xrightarrow{\alpha} {}^{234}_{90}\text{Th} \xrightarrow{\beta^-} {}^{234}_{91}\text{Pa} \xrightarrow{\beta^-} {}^{234}_{92}\text{U}$$
$$^{211}_{82}\text{Pb} \xrightarrow{\beta^-} {}^{211}_{83}\text{Bi} \xrightarrow{\alpha} {}^{207}_{81}\text{Tl} \xrightarrow{\beta^-} {}^{207}_{82}\text{Pb}$$
$$\downarrow \xrightarrow{\beta^-} {}^{211}_{84}\text{Pa} \xrightarrow{\alpha} \uparrow$$

2. $^{238}_{92}$ U decays to Pb through the successive emission of 6 electrons and 8 *a*-particles. What is the mass number of Pb isotope? What is the total energy evolved in the decay? Given:

$M(^{238}\text{U})$ = 238.050786 amu

M(Pb) = 205.9744550 amu

$M(a)$ = 4.002603 amu

$M(e) = 5.486 \square 10^{-4}$ amu

Solution: Number of nucleons in 8 *a*-particles = 8 □ 4 = 32

Therefore, the mass of Pb = 238 – 32 = 206

or

The daughter nucleus is $^{206}_{82}$ Pb

Total mass of products = $M(^{206}\text{Pb}) + 8M(a) + 6M(e)$

$= 205.974455 + 8 \square 4.002603 + 6 \square 5.486 \square 10^{-4}$

$= 237.9985706$ amu

Therefore, decrease in mass = 238.050786 – 237.9985706

= 0.0522154 amu

or

Energy evolved = 0.0522154 □ 931.47

= 48.6 MeV

Unsolved Problems

1. Complete the following reactions by putting appropriate particle on the arrow

$$^{232}_{90}\text{Th} \rightarrow {}^{228}_{88}\text{Ra} \rightarrow {}^{228}_{89}\text{Ac} \rightarrow {}^{228}_{90}\text{Th}$$

$$^{214}_{82}\text{Pb} \rightarrow {}^{214}_{83}\text{Bi} \rightarrow {}^{213}_{84}\text{Po} \rightarrow {}^{210}_{82}\text{Pb} \rightarrow {}^{210}_{83}\text{Bi} \rightarrow {}^{210}_{84}\text{Po} \rightarrow {}^{206}_{82}\text{Pb}$$

$$\downarrow\rightarrow {}^{210}_{81}\text{Tl} \rightarrow\downarrow$$

2. $^{237}_{93}$Np decays to a nucleus after 7 a-emissions and 4 b^--emissions. What is the resultant nucleus and how much energy is evolved in this process? Given:

$M(^{237}\text{Np}) = 237.048169$ amu

M(final nucleus) = 208.980388 amu

$M(a) = 4.002603$ amu

$M(e) = 5.486 \square 10^{-4}$ amu [**Ans.** ^{209}Bi, 44.12 MeV]

Section 3.4

Solved Problems

1. In the following radioactive series, write down the atomic number and mass number of B, C, D, E.

$$^{238}_{90}\text{A} \xrightarrow{\alpha} \text{B} \xrightarrow{\beta} \text{C} \xrightarrow{\alpha} \text{D} \xrightarrow{\beta} \text{E}$$

Solution:

$$^{238}_{90}\text{A} \xrightarrow{\alpha} {}^{234}_{88}B \xrightarrow{\beta} {}^{234}_{89}\text{C} \xrightarrow{\alpha} {}^{230}_{87}\text{D} \xrightarrow{\beta} {}^{230}_{88}\text{E}$$

2. ^{226}Ra decays to ^{222}Rn with a half-life of 1620 years. ^{222}Rn decays to ^{218}Po with a shorter half-life of 3.82 days. Starting with an initially pure sample of ^{226}Ra, find the number of ^{222}Rn half-lives that have elapsed when the ^{222}Rn reaches 99% of its equilibrium concentration.

Solution: We have from Eq. (3.24)

$$N_b(t) = N_a \frac{\lambda_a}{\lambda_b}[1 - e^{-\lambda_b t}]$$

and from Eq. (3.25)

$$N_b(\infty) = N_a \frac{\lambda_a}{\lambda_b}$$

Therefore,

$$\frac{N_b(t)}{N_b(\infty)} = 1 - e^{-\lambda_b t} = \frac{99}{100}$$

Therefore,

$$e^{-\lambda_b t} = \frac{1}{100}$$

or

$$t = \frac{1}{\lambda_b}\log_e(100)$$

$$= \frac{T_{1/2}}{\log_e(2)}\log_e(100)$$

$$\frac{t}{T_{1/2}} = \frac{\log_e(100)}{\log_e(2)} = 6.64$$

Therefore, number of half-lives elapsed is 6.64.

3. ^{212}Bi decays to ^{212}Ra by b^--emission in 66% of the disintegrations and to ^{208}Th by
a-emission in 34% of the cases. The total half-value period is 60.5 minutes. Find the decay constants for b^-, a and total emission.

Solution: Suppose *lb* and *la* be the decay constants for *b*- and *a*-decay respectively. The total decay constant $l_T = lb + la$.

In this case, $\lambda_\beta = \frac{66}{100}\lambda_T$ and $\lambda_\alpha = \frac{34}{100}\lambda_T$

Total half-value period = 60.5 min = 60.5 × 60 = 3630 s

Therefore,

$$\lambda_T = \frac{0.693}{3630} = 1.909 \times 10^{-4}\ \text{s}^{-1}.$$

$$\lambda_\beta = \frac{66}{100} \times 1.909 \times 10^{-4} = 1.24 \times 10^{-4}\ \text{s}^{-1}.$$

$$\lambda_\alpha = \frac{34}{100} \times 1.909 \times 10^{-4} = 6.49 \times 10^{-5}\ \text{s}^{-1}.$$

4. The atomic ratio between $^{238}_{92}$U and $^{234}_{92}$U in a mineral sample is found to be 1.8 □ 10^4. The half-life of $^{238}_{92}$U is 2.5 □ 10^5 years. Find the half-life of $^{234}_{92}$U.

Solution: Let number of atoms of $^{238}_{92}$U is N_{238} and the number of atoms of $^{234}_{92}$U is N_{234}. Let half-life of $^{238}_{92}$U be T_{238} and that of $^{234}_{92}$U be T_{234}. Using Eq. (3.27), we have

$$\frac{N_{238}}{N_{234}} = \frac{T_{238}}{T_{234}}$$

Given:

$$\frac{N_{238}}{N_{234}} = 1.8 \times 10^4$$

and

$$T_{234} = 2.5 \times 10^4 \text{ years}$$

Therefore, $$\frac{T_{238}}{2.5 \times 10^5} = 1.8 \times 10^4$$

or

$$T_{238} = 4.5 \times 10^9 \text{ years}$$

Unsolved Problems

1. The half-life of a radionuclide A is 10 min and that of its daughter B is 5 min. The daughter of B, i.e. C is stable. Calculate the number of atoms of A, B and C at an interval of 5 min up to 50 min. Given,

initially at $t = 0$, number of atoms of A equal to 10^{10}; B and C equal to zero.

2. Repeat Problem 1 taking initial activity (10 mCi) instead of number of atoms.

3. A radioactive series is as under

$$^{140}\text{Ba} \xrightarrow{12.8\text{d}} {}^{140}\text{La} \xrightarrow{40.2\text{h}}$$

Initially given 10,000 disintegrations per second (dps) of ^{140}Ba, no dps for ^{140}La. Calculate (i) dps of ^{140}Ba after 30 days and (ii) dps of ^{140}La after 12 hours and after 30 days. [**Ans.** (i) 1970, (ii) 241, 296]

4. Secular equilibrium exists between ^{226}Ra and ^{222}Rn. The half-lives of the two are respectively 1.62 □ 10^3 years and 3.82 days. Calculate the weight of ^{222}Rn, which is in secular equilibrium with 10 g of ^{226}Ra.
[**Ans.** □ 63 □g]

Section 3.5

Solved Problems

1. Are the following decays possible?

(i) $^{232}\text{U} \rightarrow {}^{231}\text{U} + {}^{1}n$

(ii) $^{232}\text{U} \rightarrow {}^{231}\text{Pa} + p$

(iii) $^{239}\text{Pu} \rightarrow {}^{235}\text{U} + {}^{4}\text{He}$

Given:

$M(^{239}\text{Pu}) = 239.052158$ amu

$M(^{235}\text{U}) = 235.043925$ amu

$M(^{231}\text{U}) = 231.036270$ amu

$M(^{231}\text{Pa}) = 231.035880$ amu

$M(^{4}\text{He}) = 4.002603$ amu

$M(^{1}n) = 1.008665$ amu

$M(p) = 1.007825$ amu

Solution:

(i) $^{232}\text{U} \rightarrow {}^{231}\text{U} + {}^{1}n$

This decay is possible if the total mass of the products is smaller than that of ^{232}U.

Total mass of the products = 231.036270 + 1.008665 = 232.044935 amu.

So, the total mass of the products is greater than that of the mass of ^{232}U. Hence, this decay is not possible.

(ii) $^{232}\text{U} \rightarrow {}^{231}\text{Pa} + p$

In this case, total mass of the products = 231.035880 + 1.007825 = 232.043705 amu.

So, the total mass of the products is greater than that of the mass of ^{232}U. Hence, this decay is not possible.

(iii) $^{239}\text{Pu} \rightarrow {}^{235}\text{U} + {}^{4}\text{He}$

In this case, total mass of the products = 235.043925 + 4.002603 = 239.046528 amu.

So, the total mass of the products is smaller than that of the mass of ^{239}Pu. Hence, this decay is possible.

2. Calculate the kinetic energy of *a*-particles in the following decay.

$$^{239}\text{Pu} \rightarrow {}^{235}\text{U} + {}^{4}\text{He}$$

Given:

$$M(^{239}\text{Pu}) = 239.052158 \text{ amu}$$
$$M(^{235}\text{U}) = 235.043925 \text{ amu}$$
$$M(^{4}\text{He}) = 4.002603 \text{ amu}$$

Solution: $^{239}\text{Pu} \rightarrow {}^{235}\text{U} + {}^{4}\text{He}$

The total disintegration energy Q is the difference in masses of the initial nucleus and the final products, or

$$Q = M(^{239}\text{Pu}) - M(^{235}\text{U}) - M(^{4}\text{He})$$
$$= 239.052158 - 235.043925 - 4.002603$$
$$= 0.00563 \text{ amu}$$
$$= 0.00563 \times 931.47 = 5.244 \text{ MeV}$$

Now, from Eq. (3.32)

$$Ka = Q^{\frac{A-4}{A}}$$

$$= 5.244 \square^{\frac{237}{241}}$$

$$= 5.16 \text{ MeV}$$

Therefore, the kinetic energy of *a*-particle = 5.16 MeV

3. Calculate the height of potential barrier faced by an *a*-particle inside the $^{226}_{88}$Ra nucleus.

Solution: The height of Coulomb barrier is given by the relation

$$B = \frac{2(Z-2)e^2}{4\pi\varepsilon_0 R}$$

$$= \frac{2(Z-2)e^2}{4\pi\varepsilon_0 R_0 A^{1/3}}$$

$$= \frac{8.98\times10^9 \times 2\times 86\times (1.6\times10^{-19})^2}{1.3\times10^{-15}\times 226^{1/3}}$$

$$= 4.9934\times10^{-12} \text{ J}$$

$$= 31.2 \text{ MeV}$$

4. Rutherford bombarded 7.7 MeV *a*-particles from ^{214}Po on ^{14}N to initiate the nuclear reaction ^{14}N(a, p)^{17}O. Find the height of Coulomb barrier faced by *a*-particles.

Solution: Height of Coulomb barrier is given by the relation

$$V_C = \frac{Z_1 Z_2 e^2}{4pe_0(R_1 + R_2)}$$

where Z_1 is the charge of the incident projectile, Z_2 is the charge of the target. R_1 and R_2 are the radii of the projectile and target nuclei respectively. We have $R = R_0 A^{1/3}$, therefore,

$$V_C = \frac{Z_1 Z_2 e^2}{4pe_0 R_0(A_1^{1/3} + A_2^{1/3})}$$

Substituting various values, we get

$$V_c = \frac{8.98\times10^{9}\times2\times7\times(1.6\times10^{-19})^2}{1.5\times10^{-15}\times(4^{1/3}+14^{1/3})}$$

$$= 5.367 \square 10^{-13} \text{ J}$$

$$= 3.35 \text{ MeV}$$

5. Find the kinetic energy required by a proton to penetrate Coulomb barrier of a hydrogen nucleus.

Solution: When a proton is bombarded on hydrogen nucleus, the Coulomb barrier E_b is given by

$$E_b = \frac{Z_1 Z_2 e^2}{r}$$

Here $Z_1 = Z_2 = 1$ and $r = 1.2$ fm

$$E_b = \frac{e^2}{r} = \frac{\hbar c}{r}\times\frac{e^2}{\hbar c}$$

or

$$E_b = \frac{197}{1.2}\times\frac{1}{137}.$$

$$= 1.2 \text{ MeV}$$

Unsolved Problems

1. Find the speeds of a-particles with energies 1 MeV and 1000 MeV.

[**Ans.** $6.95 \square 10^6$ m/s, $2.2 \square 10^8$ m/s]

2. The total energy librated in the a-decay of $^{226}_{88}$Ra is 4.87 MeV.

(i) Identify the daughter nucleus.

(ii) Find the kinetic energy of a-particle.

(iii) Find the recoil energy of the nucleus. [**Ans.** (i) $^{222}_{86}$Rn (ii) 4.78 MeV (iii) 0.09 MeV)]

3. $^{210}_{84}$Po decays by emitting a 5.30 MeV a-particle. Find the disintegration energy for this decay. Identify the daughter nucleus and find its atomic mass. Given:

$M(^{210}\text{Po}) = 209.982876$ amu

$M(^{4}\text{He}) = 4.002603$ amu [**Ans.** 5.40 MeV, $^{206}_{82}$ Pb, 205.974476 amu]

4. Determine the kinetic energy of *a*-particle emitted in an *a*-decay in terms of *Q*-value of the reaction

$$^{A}_{Z}\text{P} \rightarrow ^{A-4}_{Z-2}\text{D} + ^{4}_{2}\text{He}$$

5. Find the height of Coulomb barrier, when $^{208}_{82}$ Pb ions are bombarded on $^{208}_{82}$ Pb target. [**Ans.** 596.1 MeV]

6. Calculate the barrier height for an *a*-particle inside the $^{238}_{92}$ U nucleus. [**Ans.** 32.2 MeV]

7. Calculate the height of potential barrier faced by an *a*-particle inside the $^{252}_{98}$ Cf nucleus. [**Ans.** 33.6 MeV]

8. Find the energy released in the *a*-decay of ^{238}U

$$^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ^{4}_{2}\text{He}$$

Given $m_{\text{U}} = 238.050786$ amu. $m_{\text{TH}} = 234.043583$ amu and $m_{\text{He}} = 4.002603$ amu. [**Ans.** 4.28 MeV]

9. The straggling of heavy ions at low energy is mostly a consequence of

(a) finite momentum

(b) fluctuating state of ionization

(c) multiple scattering [**Ans.** (c) Multiple scattering changes the ions' direction of motion. This makes them straggle. Thus, straggling is due to multiple scattering.]

10. ^{212}Po decays by *a*-particle emission. Energy of these *a*-particles is 8.787 MeV. Calculate the velocity of these *a*-particles. [**Ans.** 2.06 □ 10^{7} m/s]

Section 3.6

Solved Problems

1. ^{14}C decays by b^--emission. The end point energy of b^--particles in this decay is

   0.156 MeV. Given the mass of ^{14}C = 14.007685 amu, find the mass of daughter nucleus.

Solution: The given decay is

$$^{14}C \rightarrow {}^{14}N + \beta^- + \bar{\nu}_e$$

For b^--emission with energy equal to end point energy, the energy carried by $\bar{n}_e$ = 0 MeV and mass of $\bar{n}_e$ □ 0 MeV.

Now,

$$M(^{14}N) = M(^{14}C) - \text{K.E. of } b^- \text{ (in amu)}$$

$$= 14.007685 - \frac{0.156}{931.47}$$

$$= 14.007517 \text{ amu}$$

Therefore, the mass of ^{14}N = 14.007517 amu

2. Find whether ^7Be is stable or not. If not, find the decay mode. Given:

$$M(^7Be) = 7.016928 \text{ amu}$$
$$M(^7Li) = 7.0169003 \text{ amu}$$
$$M(^7B) = 7.029920 \text{ amu}$$

Solution: For b^--decay, we have the following equation:

$$^7Be \rightarrow {}^7B + \beta^- + \bar{\nu}_e$$

From the given masses, it is clear that the mass of daughter nucleus is larger than the parent nucleus, so b^--decay of ^7Be is not possible.

For b^+-decay, we have the following equation:

$$^7Be \text{ □ } {}^7Li + b^+ + n_e$$

The condition for b^+-decay is that the mass of the parent nucleus should be greater than the daughter mass by $2m_ec^2$.

Mass of the daughter nucleus = 7.016003 amu

Mass $2m_e c^2$ and parent nucleus = $\frac{1.022}{931.47}$ + 7.016003 = 7.017100 amu

This mass 7.017100 amu is greater than the parent mass of ^{7}Be, hence this decay mode is also not possible.
For electron capture, we have the following equation:

$$^7_4\text{Be} + ^0_{-1}e \rightarrow ^7_3\text{Li}$$

This decay mode is possible as mass of ^{7}Be is greater than the mass of ^{7}Li.

3. A reservoir contains 20,000 metric tons (1 metric ton = 1000 kg) of water. If the proton mean life is 10^{32} years, how many decays you expect to observe in one year? Assume that your detector is 50% efficient and free protons and those bound in nuclei have same decay probability.

Solution: Weight of water in the reservoir = 2×10^3 metric tons = 2×10^7 kg. Each H_2O molecule contains 10 protons and its molecular weight is 18.

Therefore, 18 kg of water contains $10 \times 6.023 \times 10^{26}$ protons.

2×10^7 kg of water have $\frac{10}{18} \times 6.023 \times 10^{26} \times 2 \times 10^7$ protons

$= 6.70 \times 10^{33}$ protons

Therefore, the expected number of proton decays observed per year

$= \frac{6.70 \times 10^{33}}{t_p} = \frac{6.70 \times 10^{33}}{10^{32}}$

= 67.0 decays/year

Since the efficiency of the detector is 50%, therefore the number of proton decays observed per year = 67.0×0.5 = 33.5 decays/year

Unsolved Problems

1. Find whether $^{64}_{29}$Cu is stable or not. If not, find the decay mode. Given:

$M(^{64}\text{Cu}) = 63.929766$ amu

$M(^{64}\text{Zn}) = 63.929145$ amu

$M(^{64}\text{Ni}) = 63.927968$ amu

2. Calculate the maximum energy of electrons emitted in b^--decay of ^{12}B. Given:

$M(^{12}\text{B}) = 12.014353$ amu

$M(^{12}\text{C}) = 12.0000$ amu [**Ans.** 12.9 MeV]

3. Sixty milligrams of naturally occurring potassium shows b^--activity of 1.9 disintegrations per second. Isotope ^{40}K present in natural potassium is responsible for this activity. ^{40}K is 0.012% of the natural potassium. Find the half-life of ^{40}K. [**Ans.** 1.25 □ 10^9 years]

4. One student claims that he has observed ^{16}N to decay by b^--particle emission. Other student claims that ^{16}N decays by a-emission. Which student is correct? Given:

$M(^{16}\text{N}) = 16.006100$ amu, $M(^{16}\text{O}) = 15.9945915$ amu

$M(^{12}\text{B}) = 12.014353$ amu, $M(^{4}m_e) = 4.002603$ amu

$M(b^-) = 0.0005486$ amu. [**Ans.** first student is correct]

5. A radioisotope emits b-particles. The counts due to b-activity of a freshly prepared sample of this element is measured with a GM counter, the following data are obtained.

Time (minute)	*Counts*
0	1080
20	732
40	509
70	300
100	175
130	101

160	59
190	37

(i) Plot the logarithm of the counts against time.
(ii) Determine the disintegration constant from the slope of the curve.
(iii) Find the half-life of the element. [**Ans.** (ii) 2.94 □ 10^{-4} s^{-1}, (iii) 39.2 min]

6. ^{64}Cu nucleus decays by b^-, b^+ and electron capture with a half-life of ~13 hrs. Write down the nuclear reactions for these three processes and identify the daughter nucleus in each case. [**Ans.** ^{64}Zn, ^{64}Ni, ^{64}Ni]

7. ^{126}I nuclei decay by negatron emission, positron emission and electron capture. Write the decay equation for each mode of decay and identify the daughter nuclide.
[**Ans.** $^{126}_{53}\mathrm{I} \rightarrow {}^{126}_{54}\mathrm{Xe} + b^{?} + \bar{\nu}$, $^{126}_{53}\mathrm{I} \rightarrow {}^{126}_{52}\mathrm{Te} + b^{+} + \nu$, $^{126}_{53}\mathrm{I} + e^{-} \rightarrow {}^{126}_{52}\mathrm{Te} + \nu$]

8. Out of the two nuclei ^{64}Ni and ^{64}Cu, which can undergo b^--decay? Given the masses of ^{64}Ni = 63.927966 amu, ^{64}Cu = 63.929764 amu, ^{64}Zn = 63.929142 amu and $m_e = 5.4858 \times 10^{-4}$ amu. [**Ans.** ^{64}Ni decay not possible, ^{64}Cu decay possible]

Section 3.7

Solved Problems

1. Refer to Figure 3.24, the first excited state of ^{60}Ni is at 1332 keV and the second is at 2505 keV. Second excited state decays to first state, which in turn decay to ground state. Find the energies of the two *g*-rays.

Solution: When nucleus makes a transition from 2505 keV state to 1332 keV state, energy of the *g*-rays emitted = 2505 – 1332 = 1173 keV

When nucleus makes a transition from first excited state to ground state, energy of the *g*-rays emitted = 1332 – 0 = 1332 keV.

2. ^{137}Cs decays by b^- emission as shown in Figure 3.23. Find the energies of K and L shell conversion electrons. The binding energies of K and L shell electrons in ^{137}Ba is 37.4 and 6.0 keV respectively.

Solution: Energy of the conversion electron $E_c = Eg - I_{be}$.

When conversion takes place from K shell

$$E_{cK} = Eg - I_{beK} = 662 - 37.4 = 624.6 \text{ keV}$$

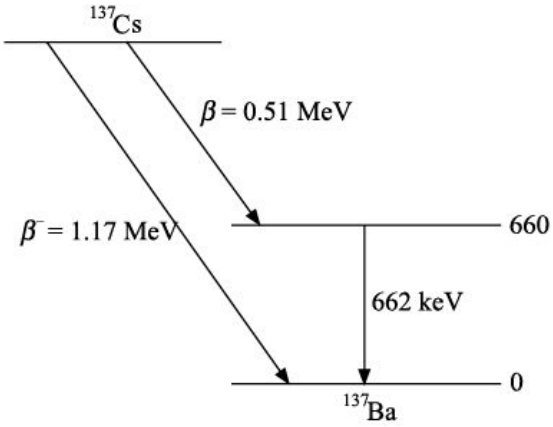
When conversion takes place from K shell

$$0E_{cL} = Eg - I_{beL} = 662 - 6.0 = 656 \text{ keV}$$

3. ^{137}Cs decays by emitting two groups of b^--particles of energies 0.51 and 1.17 MeV. The daughter nucleus ^{137}Ba also emits a g-ray of energy 662 keV. Use this information to sketch the decay scheme. You may assume that the most energetic b^--particle leaves the daughter nucleus in its ground state.

Solution: While constructing this decay scheme, we allow a possible error of about 1 keV. As the most energetic b^--particle leaves the daughter nucleus in the ground state, therefore the 1.17 MeVb-particle populates the ground state of the daughter nucleus. The energy of the ground state is assumed to be 0. b-particle of energy 0.51 MeV leaves the nucleus in the first excited state. The energy of the first excited state will be 1.17 – 0.51
= 0.66 MeV or 660 keV.

The decay scheme constructed is as shown.



4. The daughter nucleus of a given *a* emitter has several excited states and therefore the kinetic energy of an emitted particle can have one of several possible values. For a particular heavy nucleus, the energies (in MeV) of *a*-particles are:

5.545, 5.513, 5.486, 5.469, 5.443, 5.417, 5.389.

The daughter nucleus formed after *a* decay emits several *g*-rays with the following energies (in keV): 26, 33, 43, 56, 60, 99, 103, 125.

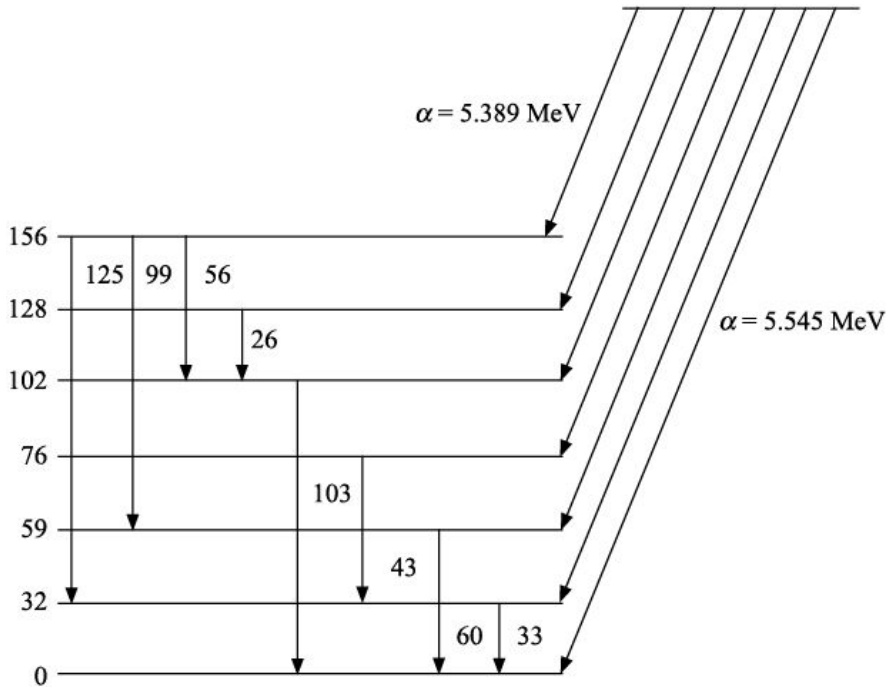
Use this data to draw a decay scheme indicating the energy levels and marking the *g*-ray transitions. You may assume that the most energetic *a*-particle leaves the daughter nucleus in its ground state.

Solution: While constructing this decay scheme, we allow a possible error of about 1 keV.

As the most energetic *a*-particle leaves the daughter nucleus in the ground state, therefore the 5.545 MeV *a*-particle populates the ground state of the daughter nucleus. The energy of the ground state is assumed to be 0. *a*-particle of energy 5.513 MeV leaves the nucleus in the first excited state. The energy of the first excited state will be 5.545 – 5.513 = 0.032 MeV or 32 keV. Similarly, 5.486 MeV *a*-particle leaves the daughter nucleus in the second excited state, whose energy is 5.545 – 5.486 = 0.059 MeV or 59 keV. In this way, we construct the level scheme of the daughter nucleus.

After constructing the level scheme, the energies of *g*-rays emitted by the daughter nucleus can be estimated by measuring the energy difference between various levels by hit and trial. If the energy difference between any two levels matches with the *g*-ray energy in the given list, then most probably that *g*-ray is due to transition between those two levels.

The decay scheme constructed and the position of various *g*-rays is as shown.



Unsolved Problems

1. The measured width of the *g*-ray of ^{55}Fe is 3.4 □ 10^{-8} eV. Determine the lifetime of this excited state. The relation between the width (*g*) and the lifetime (*t*) is *g* □□ *t* ≡ *h*, where *h* is Planck constant. [**Ans.** 19.36 ns]

2. The lifetime of the excited state of ^{203}Hg is 8 □ 10^{-13} s. Determine the width of the
g-ray emitted when the nucleus goes to the ground state. The relation between the width (*g*) and the lifetime (*t*) is *g* □ *t* ≡ *h*, where *h* is Planck constant.

[**Ans.** 5.17 □ 10^{-3} eV]

3. ^{226}Ra decays by emitting two groups of *a*-particles of energies 4.785 and 4.602 MeV. The daughter nucleus ^{222}Rn also emits a *g*-ray of energy 186 keV. Use this information to sketch the decay scheme. You may assume that the most energetic *a*-particle leaves the daughter nucleus in its ground state.

4. For a nuclide *X* with the decay scheme how many *g*-rays are emitted per 100 disintegrations of *X* if the coefficient for internal conversion is 0.25?

Section 3.8

Solved Problem

1. An isotope of element 105 was prepared by bombarding a $^{249}_{98}$ Cf target with $^{15}_{7}$ N, $^{249}_{98}$ Cf emits four neutrons. Write the equation for the nuclear reaction.

Solution: The reaction is

$$^{249}_{98}\text{Cf} + ^{15}_{7}\text{N} \rightarrow ^{260}_{105}\text{X} + 4\,^{1}_{0}n$$

Unsolved Problems

1. $^{243}_{97}$Bk is prepared by bombarding ^{14}N on ^{238}U. During this reaction, an *a*-particle and 5 neutrons are ejected. Complete this reaction. [**Ans.** $^{14}_{7}\text{N} + ^{238}_{92}\text{U} \rightarrow ^{243}_{97}\text{Bk} + ^{4}_{2}\text{He} + 5\,^{1}_{0}n$]

2. Element Astatine (At) is prepared by bombarding ^{12}C on $^{197}_{79}$Au. During this process, 4 neutrons are emitted. What is the atomic number of Astatine? Complete the reaction. [**Ans.** 85; $^{12}_{6}\text{C} + ^{197}_{79}\text{Au} \rightarrow ^{205}_{85}\text{At} + 4\,^{1}_{0}n$]

Section 3.9

Solved Problems

1. In a uranium material, lead is found predominantly in the form of isotope 206. The mineral contains 0.093 g of lead in 1 g of uranium. Calculate the age of mineral. Half-life of $^{238}_{92}$ U = 4.5 □ 10^9 years and all other elements of this series have very short-lives.

Solution: Let N_0 and N be the number of ^{238}U nuclei at time $t = 0$ and at time t. The difference $N_0 - N$ is the number of nuclei of ^{206}Pb at time t.

Now, $N = N_0 e^{-lt}$, where l is decay constant of ^{238}U

Now, $t_{1/2} = 4.5 \square 10^9$ years

So, $$l = \frac{0.6931}{4.5 \times 10^9} \text{ years}^{-1}$$

$$= 1.54 \square 10^{10} \text{ years}^{-1}$$

Also, $$N_0 - N = N_1 = N_0(1 - e^{-}l^t)$$

$$\frac{\text{Number of } ^{206}\text{Pb atoms at time } t}{\text{Number of } ^{238}\text{U atoms at time } t} = \frac{N_1}{N}$$

$$= \frac{1 - e^{\lambda t}}{e^{-\lambda t}}$$

$$= e^{\lambda t} - 1$$

So,

$$e^{\lambda t} = 1 + \frac{N_1}{N}$$

or $$\lambda t = \log_e\left(1 + \frac{N_1}{N}\right)$$

So, $$t = \frac{1}{\lambda}\log_e\left(1 + \frac{N_1}{N}\right)$$

Number of nuclei in 238 g of ^{238}U = $6.023 \square 10^{23}$

Number of nuclei in 1 g of ^{238}U (N) = $6.023 \square 10^{23} \square \frac{1}{238}$

Number of nuclei in 206 g of ^{206}Pb = $6.023 \square 10^{23}$

Number of nuclei in 0.093 g of ^{206}Pb (N_1) = $6.023 \square 10^{23} \square \frac{0.093}{206}$

Now,

$$\frac{N_1}{N} = 6.023 \times 10^{23} \times \frac{0.093}{206} \times \frac{238}{6.023 \times 10^{23}}$$

$$= 0.1074$$

Therefore,

$$t = \frac{1}{\lambda}\log_e\left(1 + \frac{N_1}{N}\right)$$

$$= \frac{1}{1.54 \times 10^{-10}}\log_e(1 + 0.1074)$$

or

Age of the mineral $t = 0.6626 \square 10^9$ years

2. A sample of carbon from an ancient wooden boat piece gives 5 count/min/g of carbon due to ^{14}C present in it. If freshly cut wooden piece gives 16 count/min, what is the age of the boat? Half-life of ^{14}C = 5760 years.

Solution: We have

$N = N_0 e^{-l^t}$

or

$$e^{\lambda t} = \frac{N_0}{N} \Rightarrow \lambda t = \log_e \frac{N_0}{N}$$

or

$$t = \frac{1}{\lambda} \log_e \frac{N_0}{N}$$

Since $$\lambda = \frac{0.693}{T_{1/2}}$$

Therefore,

$$t = \frac{T_{1/2}}{0.693} \log_e \frac{N_0}{N}$$

Substituting various values, we get

$$t = \frac{5760}{0.693} \log_e \frac{16}{5}$$

or

$t = 9668$ years

3. (a) What type of reaction is taking place in a nuclear reactor?

(b) What is the average energy released per reaction in a reactor?

(c) Why are the reaction products radioactive?

(d) What is the necessity of a moderator in a nuclear reactor? Are light or heavy elements preferred for moderators and why?

Solution:

(a) In nuclear reactors, fission of heavy nuclei like ^{233}U, ^{235}U, ^{239}Pu, take place.

(b) In fission of heavy nuclei, about 200 MeV of energy is released per fission.

(c) In fission, some of the fission fragments are left with high excitation energy.

(d) The fission is caused by thermal neutrons (having average energy of about 0.025 eV). However, in fission, fast neutrons are produced. To slow down these fast neutrons, moderator is required. Light nuclei (like H_2, D_2, D_2O, Carbon, etc.) are good moderators as energy transferred in a single collision with light moderator is quite large.

4. A by-product of fast breeder reactor is ^{239}Pu. It can be used as fuel in nuclear reactors. It is an *a*-emitter with a half-life of 24,000 years. Consider 10 g of ^{239}Pu at $t = 0$.

(a) What is the number of nuclei at $t = 0$?

(b) What is the initial activity?

(c) For how long you need to store ^{239}Pu until it has decayed to a safe activity level of 0.1 Bq?

Solution:

(a) 239 kg of ^{239}Pu contains 6.023×10^{26} nuclei.

0.01 kg of ^{239}Pu will contain $\frac{6.023 \times 10^{26}}{239} \times 0.01$ nuclei

$$N = 2.52 \times 10^{22} \text{ nuclei}$$

(b) Half-life of ^{239}Pu = 24,000 years

Therefore, decay constant

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.6931}{24{,}000 \times 365 \times 2400 \times 3600}$$

$$= 9.158 \times 10^{-13} s^{-1}$$

Therefore,

Initial activity = $l \times N$

$= 9.158 \times 10^{-13}\ 2.52 \times 10^{22}$

$= 2.31 \times 10^{10}$ disintegrations per second

(c) We have

$$A = A_0 e^{-\lambda t}$$

which gives

$$t = \frac{1}{\lambda} \ln \frac{A_0}{A}$$

Substituting various values, we get

$$t = \frac{1}{9.158 \times 10^{-13}} \ln \frac{2.31 \times 10^{10}}{0.1}$$

or

$$t = 2.86 \square 10^{13} \text{ s} = 0.91 \square 10^{6} \text{ years}$$

5. Radioactive dating of once living objects like trees, etc. is done using the isotope

(a) ^{40}K

(b) ^{12}C

(c) ^{14}C

Solution: ^{14}C maintains a small but fixed proportion in the CO_2 present in the atmosphere. A living species also maintains the same isotopic proportion of ^{14}C. After it dies, this proportion starts changing. This provides a means of dating the time of death. ^{12}C is stable and cannot be used for dating purposes, ^{40}K has a half-life of $1.25 \square 10^{9}$ years and is also not present in atmosphere and hence cannot be used for dating purposes for once living species.

Unsolved Problems

1. A typical smoke detector used in homes has approximately 0.5 □Ci of ^{241}Am in it. How many atoms of ^{241}Am does this represent? Given: $\lambda = 0.0015 \text{ year}^{-1}$. [**Ans.** $1.05 \square 10^{4}$ atoms]

2. A sample of carbon from an ancient tree gives 3.2 count/min/g of carbon due to ^{14}C present in it. What is the approximate age of the tree? Half-life of ^{14}C = 5760 years and a freshly cut wooden piece gives 16 count/min? [**Ans.** 13,380 years]

3. Some of the carbon atoms in trees and lumber are ^{14}C. There are none among the carbon atoms in the petroleum products. Why?

4. A piece of charcoal recovered from a fire pit in an ancient camp site has average

b-activity due to ^{14}C content of charcoal as 9.2 count/min/gm. What is the age of the charcoal sample? Given the absolute activity of the ^{14}C in the wood from a living tree is independent of the species of the tree and its average value is 16.0 disintegrations per minute per gram of sample. [**Ans.** 4600 years]

5. ^{206}Pb is found in certain U ore. This indicates that the lead is of radioactive origin. What is the age of the U ore if it now contains 0.80 g of ^{206}Pb for each gram of ^{238}U? [**Ans.** 4.25 □ 10^9 years]

6. At present the abundance of the isotopes in natural uranium are 0.72% of $^{235}_{92}$ U and 99.28% of ^{238}U. What were the values at the time of formation of the earth, i.e.

4.55 □ 10^9 years ago? The half-lives of $^{235}_{92}$ U and ^{238}U are 7.13 □ 10^8 years and

4.51 □ 10^9 years respectively. [**Ans.** 23.1%, 76.9%]

7. ^{232}Th have a half-life of 1.41 □ 10^{10} years. Its successive radioactive decays lead to the stable isotope ^{208}Pb. A piece of rock contains 3.65 g of ^{232}Th and 1.25 g of ^{208}Pb. Assume that all the ^{208}Pb has come from the decay of ^{232}Th, what is the age of the rock as determined from the Th/Pb ratio? [**Ans.** 6.58 □ 10^9 years]

8. The ^{14}C content decreases after the death of a living system with a half-life of

5600 years. If the ^{14}C content of an old piece of wood is found to be 6.5% of that of an equivalent present-day sample, how old is the piece of wood? [**Ans.** 22,088 years]

9. ^{14}C decays with a half-life of 5600 years.

(a) What is the nature of this decay and what are the final products?

(b) Find the age of a dead tree whose radioactivity is 1/4 of that of a comparable but living young tree.
[**Ans.** $^{14}C \rightarrow {}^{14}N + e^- + \bar{\nu}_e$, 11,200 year]

10. A human body weighing 70 kg contains about 23% carbon, out of which about 10^{-10}% are ^{14}C atoms. Determine the total number of ^{14}C atoms in the body and also determine its activity. Given half-life of ^{14}C is 5715 years. [**Ans.** 8.08 □ 10^{14}, 84nCi]

11. A normal human body contains about 150 g of potassium. Determine the total number of atoms of ^{40}K present in the body. Also determine its activity. Given abundance of ^{40}K in natural potassium is 0.0117% and its half-life is 1.27 □ 10^9 years. [**Ans.** 2.71 □ 10^{20}, 127nCi]

REVIEW QUESTIONS

Short Answer Type

1. Why do the number of neutrons tend to exceed the number of protons in stable nuclei?
2. Explain the difference between X-rays and *g*-rays.
3. Explain the nuclear potential barrier for *a*-decay with suitable diagram and hence show how this decay is not possible classically.
4. In case of successive radioactive disintegration, what is the meaning of permanent equilibrium?
5. Differentiate between a nuclear decay process and a nuclear reaction.
6. A certain radioactive element disintegrates for an interval of time equal to its mean life. What fraction of element has disintegrated?
7. Discuss the factors which support the neutrino hypothesis of *b*-decay.
8. What are radioisotopes? Discuss their three uses.
9. Describe the law of absorption of *g*-rays in matter.
10. Explain Geiger–Nuttal law.
11. *a*-rays and *g*-rays can have the same energy. What is the difference between them?
12. What are internal conversion and electron capture processes?
13. What is neutrino? How does its emission explain *b*-spectrum?

14. What is radioactive dating?
15. Why is *b*-spectrum continuous?
16. What does the term nuclear winter mean?
17. How are atomic number and mass number changes during *a*-, *b*- and *g*-decays?
18. What is a radioactive tracer? Give its two applications.
19. What are the important features of *b*-ray spectra? What is end point energy?
20. What do you understand by internal conversion?
21. How is the age of earth calculated from the radioactive decay?
22. Define two units to measure the intensity of radioactivity.
23. Define decay constant.
24. What is the principle of radioactive dating?
25. Although nucleus is positively charged, how will you explain *b*-rays coming out of it?
26. What do you mean by end point energy of a *b*-particle?
27. List some similarities and differences between the properties of photons and neutrinos.
28. Explain the concept of parity violation in *b*-decay.
29. Name three types of radiations which are emitted by radioactive substances.
30. Why is energy of *a*-particles discrete in *a*-decay?
31. Explain the phenomenon of *a*-decay on the basis of uncertainty principle.
32. What happens to the atomic number and mass number of a nucleus when it
(i) emits electron,
(ii) emits positron,
(iii) captures electron.
33. What do you mean by radioactivity?
34. Explain the different modes of radioactive disintegration.
35. What is the S.I. unit of radioactivity? Define it.
36. Define average life of a radioactive material.
37. Why is neptunium series not observed in nature?
38. Is *a*-spectrum a line spectrum or continuous spectrum? Explain.
39. Write few important properties of *a*-particles.

40. Define the range of *a*-particles.
41. What is internal conversion process?
42. Define internal conversion coefficient.
43. What is induced radioactivity? Give one example.

Long Answer Type

1. What is *b*-decay? Explain its spectrum. What led Pauli to predict the existence of neutrino?
2. Give the *a*-decay scheme and discuss the Geiger–Nuttal law completely.
3. What is *b*-decay? Describe neutrino hypothesis of *b*-decay. What is the evidence for existence of neutrino?
4. Discuss the theory of successive decay of radioactive substances and obtain the conditions for transient and secular equilibrium.
5. What is neutrino hypothesis? What is the need and significance of this hypothesis?
6. Explain how quantum mechanics helps to explain the *a*-decay process. Also show mathematically that *a*-particle takes up most of the *Q*-value as its kinetic energy.
7. Compare the positron emission and electron capture.
8. Explain the Geiger–Nuttal law in terms of the range of *a*-particles and also in terms of the energy of *a*-particles. Also compare the graphs obtained for different radioactive series in accordance with this law.
9. Describe internal conversion process.
10. What do you understand by successive disintegration and what is the condition for permanent equilibrium?
11. Describe the important features of *b*-ray spectrum emitted by radioactive nuclides. Discuss the arguments which led Pauli to postulate the existence of neutrino. Outline the neutrino theory of *b*-decay. How does it explain the continuous energy spectrum in *b*-decay?
12. What is the range of *a*-particles? Discuss the Geiger–Nuttal law, giving its importance.
13. What is radioactivity? Deduce statistical law of radioactivity. What is decay constant, half-life and average life of radioactive substances?
14. Give the theory of *a*-decay.

15. Briefly explain the phenomena:
(i) Electron capture.
(ii) Internal conversion.

16. What do you understand by half-life and mean life of a radioactive substance? Derive expressions for them.

17. Explain the Geiger–Nuttal law and discuss its importance.

18. Describe neutrino hypothesis of *b*-decay. What is the evidence for the existence of neutrino? Give the qualitative account of the theory of *b*-decay.

19. What do you mean by radioactive series? Explain thorium series. Why is neptunium series generally not considered?

20. Explain the theory of *b*-decay. How did it help Pauli in predicting neutrino?

21. Write short notes on:
(i) Geiger–Nuttal law.
(ii) Internal conversion.

22. Explain the term radioactive dating, radioactive series, radioactive tracers, and radioactive decay constant.

23. Give the theory of *b*-decay.

24. What is neutrino? What were the factors which led Pauli to give his hypothesis? Show that this hypothesis has been able to explain them.

25. What is *a*-decay? Discuss the energetics of *a*-decay.

26. Explain neutrino hypothesis.

27. Explain the processes of *b*-decay, positron emission and electron capture.

28. What is Geiger–Nuttal law?

29. What are the different modes of *b*-radioactivity?

30. Explain the energetics of *b*-decay.

31. Explain how neutrino hypothesis explains the process of *b*-decay.

32. Describe the salient features of *b*-decay process.

33. Describe the salient features of *b*-decay of nuclei. What is the evidence for the existence of neutrino?

34. Discuss the phenomenon of *a*-decay qualitatively.

35. State and explain the laws of radioactive disintegration. Define disintegration constant and half-life.

36. Explain qualitatively the phenomenon of *g*-decay in nuclei.

37. Explain the method of radioactive dating.
38. Discuss parity violation in *b*-decay.
39. Explain the theory of *a*-decay and derive the Geiger–Nuttal law.
40. Determine the kinetic energy of *a*-particles emitted in an *a*-decay in terms of *Q*-value reaction.
41. What are the laws of radioactive disintegration? Derive the relation $N = N_0 e^{-lt}$.
42. Show that radioactive decay is exponential in nature.
43. Prove that it will take infinite time to disintegrate whole of the radioactive material.
44. Define activity. Derive the relation $A = A_0 e^{-lt}$.
45. Define half-life of a radioactive material. Find the relation between half-life and disintegration constant.
46. What is average life of a radioactive material? Prove that the average life of a radioactive material is given by the relation $t = \frac{t_{1/2}}{0.6931}$.
47. Explain secular and transient equilibrium.
48. Define *a*-decay. Find the expression for the disintegration energy in the *a*-decay process.
49. Define the range of *a*-particles. On what factors does it depend?
50. Explain the conditions for spontaneous emission of *b*-particles.
51. Write short notes on detection of neutrino (*n*) and antineutrino ().
52. Discuss briefly the applications of radioactivity.